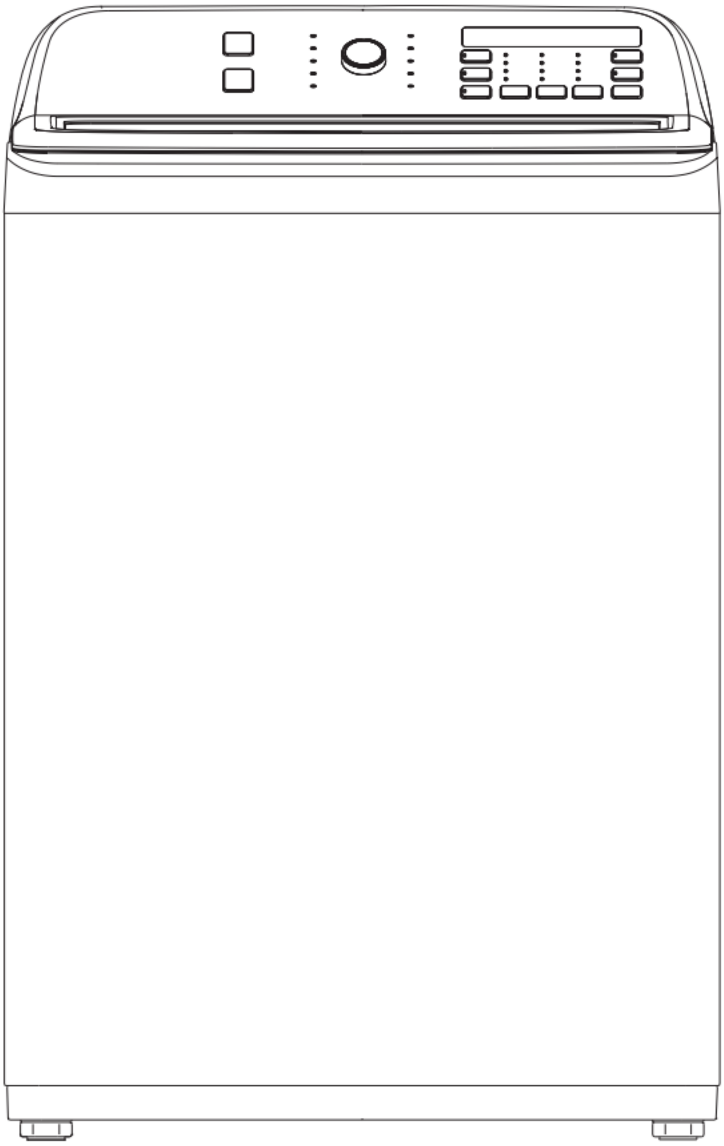
Service Manual



Model: MAV160-A2810PS/01FM-US13

Note:

Read this service manual carefully before service.

Please contact your service center if you have any problem.

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WARNING

- Unplug the power cord once you don’t need it
- Do not hold the plug by a wet hand
- Failing to unplug may cause an electric shock

IMPORTANT SAFETY NOTICE：

This service information is intended for individuals possessing adequate backgrounds of electrical, electronic and mechanical experience. Any attempt to repair this appliance may result in personal injury or property damage. The manufacturer or seller can not be responsible for the interpretation of this information, nor can it assume any liability in connection with its use.

01 PRECAUTION

NOTICE:
During the repair process, please pay attention to the following safety precautions when troubleshooting and replacing parts.

Safety Precautions

- **Use original parts**

The components of a washing machine have safety features such as non-flammability and voltage resistance. Therefore, it is essential to use the same components recommended by the manufacturer. Especially for labeled major safety components, it is necessary to only use the specified parts.

- **Grounding**

Connect the grounding wire to the outer shell plate and bury it at least 25 centimeters underground, or connect the grounding wire to the appropriate pin on a properly grounded power outlet. Do not connect the grounding wire to telephone lines, lightning rods, or gas pipelines.

Service Precautions

- **Observe Warnings**

Be sure to follow special warning and precautions that are described on part labels and in the owner's manual.

01 PRECAUTION

- **Parts Assembly and Wiring**

Be sure to use insulation material (such as tube and tape). And be sure to restore all parts and wires to their original position. Take special care to avoid contact with sharp edges.

- **Perform Safety Checks after Service**

After servicing, check to see that the screws, parts, and wiring are restored to their original positions, and
check the insulation between the external metals and the socket plug. In addition, place the washing machine
in a level position (less than 1(one) degree) to prevent vibration and noise during operations.

- **Insulation Checks**

Pull out the plug from the power receptacle, pour water into the spin tub, and then set the timer. Check to see that the resistance insulation between the terminals of the plug and the externally exposed metal is greater than 1M.

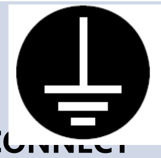
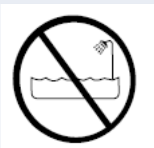
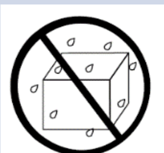
NOTICE:

When it is impossible to perform insulation check with a 500V insulation resistance tester, use other testers for inspection.

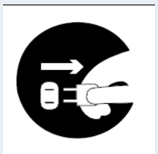
01 PRECAUTION

- Please observe the following notes for safety.
- The symbols indicate as follows.

Symbol	Meaning
 **WARNING**	Indicates possibility of death or serious injury of a repair technician and a person nearby through the misconducted work, or of a user by a defect of the product after the work performed by the technician.
 **USE EXCLUSIVE SOCKET**	• Use an exclusive A socket for the washing machine. • Otherwise, an electric shock or ignition may cause. • Sharing the same socket with other instrument causes heating of a branch socket and result in a fire.
 **REMOVE DUST**	• Wipe off dust adhered to the plug of power cord. • Dust may cause a fire.

Graphic Symbol	Meaning
 **CONNECT GROUNDING WIRE**	• Connect the grounding wire. • Failing to do so may cause an electric shock when a short circuit occurs. • Consult an electric work shop or a sales shop.
 **DO NOT USE WET PLACE**	• Do not install in a bath room or a place exposed to wind or rain. • An electric shock or a short circuit may cause a fire.
 **DO NOT SPLASH WATER**	• Do not pour or immerse electrical parts into water or liquid solution. • An electric shock or ignition may occur.

WARNING

	• Advise the customer to keep children out of the work place. • Children may be injured with a tool or a disassembled part.
	• Unplug power cord for the work such as disassembling which is not unnecessary to power on . Do not hold the plug by a wet hand. • Failing to unplug may cause an electric shock.
	• Use the specified repair parts when repairing the product. • Otherwise, a malfunction or a defect may occur. • Also, a short circuit, ignition or other danger to the customer may occur.
	• After repair, measure insulation resistance between the charging part (power cord plug) and the non-charging metallic part (ground) with an insulation resistance meter (500V). The resistance shall be 10M• or more. • Failing to check the insulation resistance may cause a short circuit, electric • shock or other diseases to the customer.
	• Do not modify the product. • An electric shock or ignition may occur.
	• Only a repair technician can disassemble and repair. • An electric shock, ignition or malfunction may cause injury.

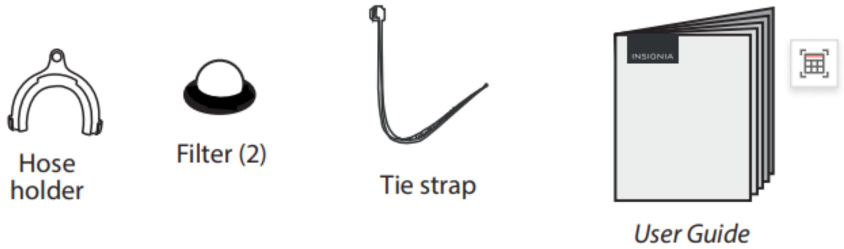
02 SPECIFICATIONS

Dimensions (H × W × D)	43 5/16 x 27 × 28 15/16 in. (110 × 68.6 × 73.5 cm)
Weight	**Net**: 130 lbs. (59 kg) **Gross**: 143.3 lbs. (65 kg)
Power	120V AC at 60Hz
Current draw	10A
Display type	LED
Display color	White
Motor	Series

02 SPECIFICATIONS

Package contents

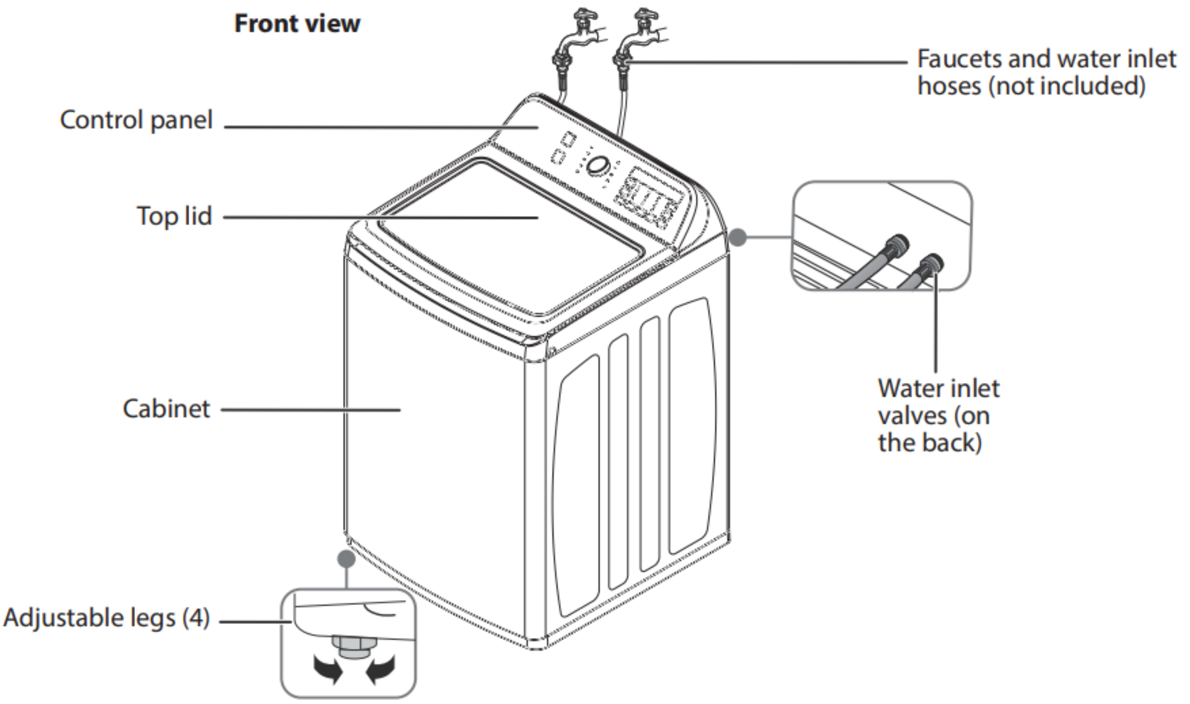
Make sure that the following items came with your washer:



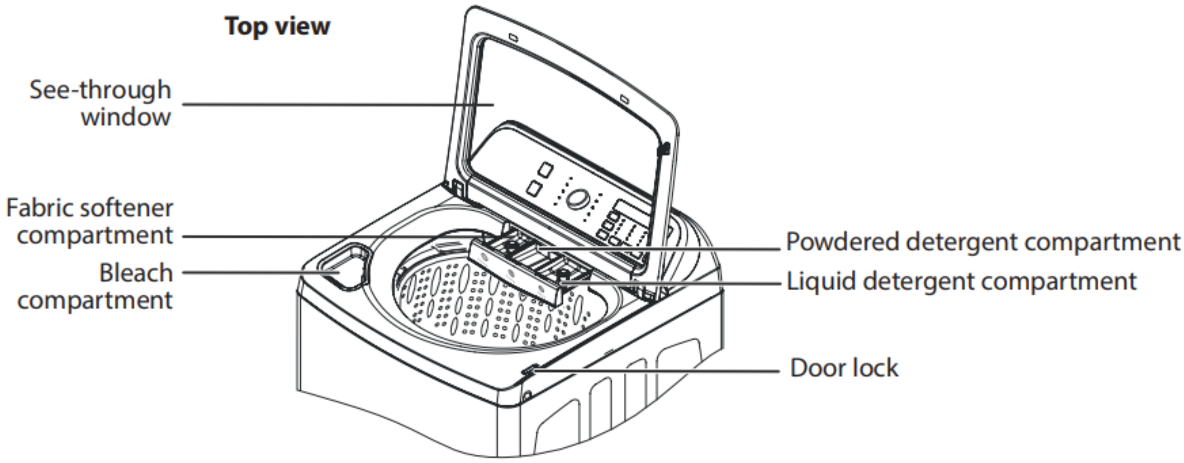
Note: Use the filters to replace the rubber washers in the water hoses (faucet ends).

Main components

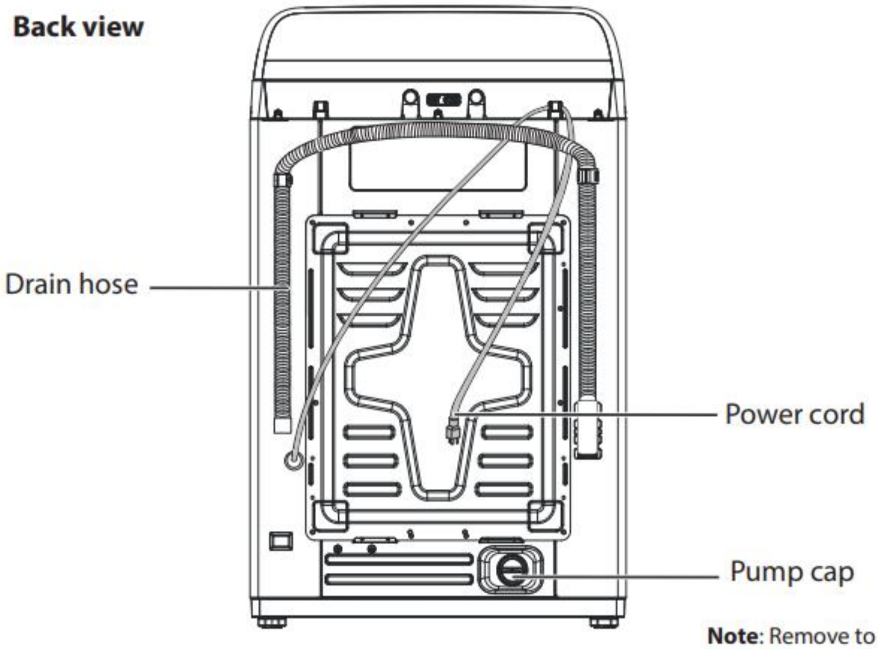
Front view



Top view



02 SPECIFICATIONS



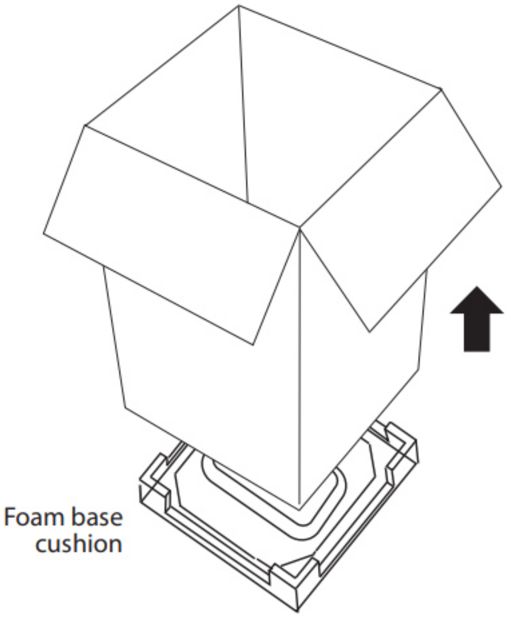
Note: Remove to drain the washer for long term storage or to remove pet hair or other insoluble items that could interfere with the washer draining properly.

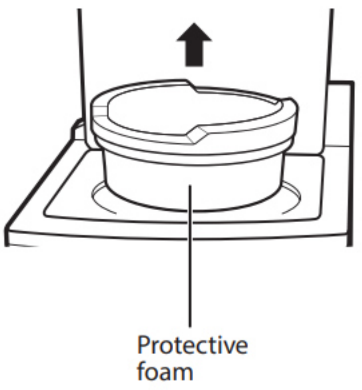
03 INSTALLATION

Unpacking your washer

WARNING:
- Packing materials can be dangerous to children. Keep all packing materials (plastic bags, polystyrene, and so on) well out of the reach of children.
- Do not operate your washer on the foam base cushion. Doing so will result in serious vibration, which could cause equipment damage or result in physical injury.

- Remove the packing box and lift your washer up and away from the foam base cushion. Open the lid of your washer to take out all accessories, BUT do not remove the protective foam until your washer is in it's final location. This will hold the tub in place while moving.





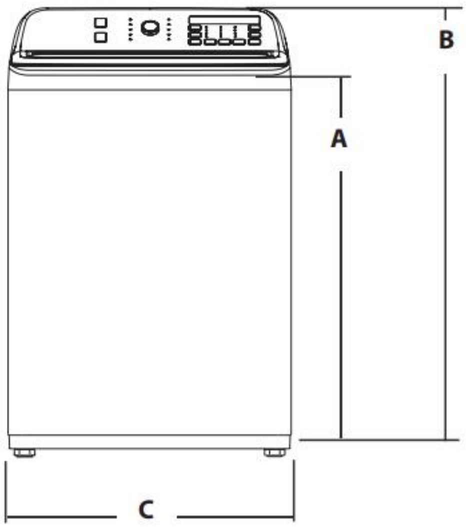
03 INSTALLATION

Choosing a location

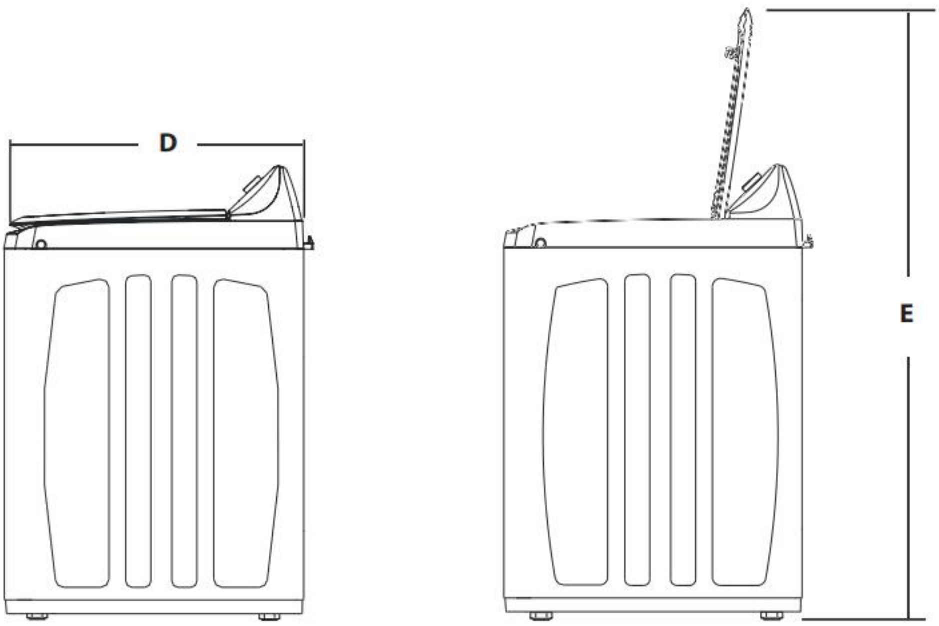
WARNING:
- Do not install your washer in areas where water may freeze, because your washer always retains some water in its water valve, pump, and hose areas. Frozen water can damage the belts, pump, hoses, and other components.
- Never install on a platform or weakly supported structure.

For best performance, you must install your washer on a solid, level floor.
Wooden floors may need to be reinforced to minimize vibration or unbalanced load situations.
Carpeting and soft tile surfaces can contribute to excessive vibration, which can cause your washer to move slightly during the spin cycle.

Washer dimensions and installation measurements



A	38 7/16 in. (97.5 cm)
B	43 5/16 in. (110 cm)
C	27 in. (68.6 cm)
D	28 15/16 in. (73.5 cm)
E	56 3/4 in. (144 cm)



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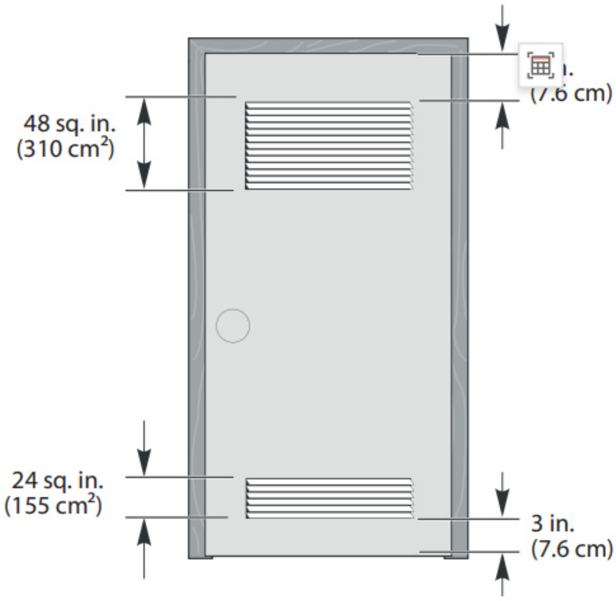
If you install your washer in an alcove or closet, the front of the closet must have two unobstructed air openings for a combined minimum total area of 72 sq. in. (465 sq. cm) with a minimum clearance of 3 in. (7.6 mm) at the top and bottom. A slatted door with equivalent space clearance is acceptable.

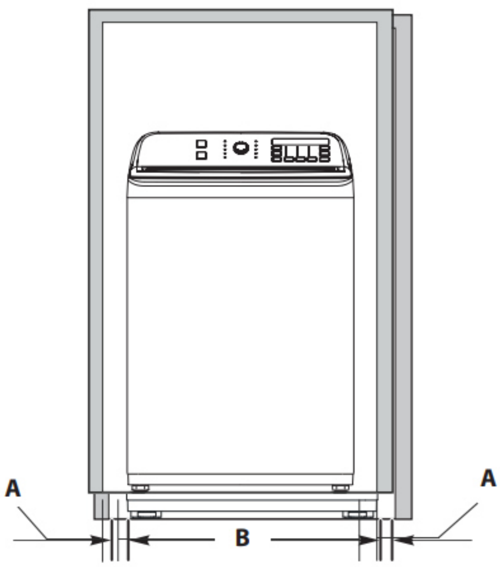


Installing in an alcove or closet

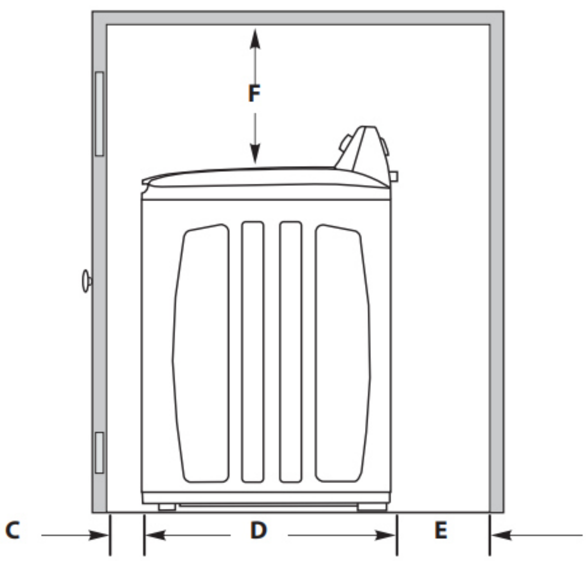
Minimum clearances between your washer and adjacent walls or other surfaces are:

- **Either side:** 1 in. (2.5 cm)
- **Rear:** 6 in. (15.2 cm)
- **Closet Front:** 2 in. (5.1 cm)
- **Top:** 24 in. (61 cm)
- **Gap between the end of the water valve and the wall:** 6 in. (15.2 cm)

Recessed area - front view



Recessed area - side view



A	1 in. (2.5 cm)
B	27 in. (68.6 cm)
C	2 in. (5.1 cm)

D	28 1/4 in. (71.5 cm)
E	6 in. (15.2 cm)
F	24 in. (61 cm)

03 INSTALLATION

Electrical requirements

WARNING: To reduce the risk of fire, electric shock, or injury to persons, read the Important Safety Instructions on page 2 before operating this appliance.

Use a 120V AC at 60Hz, 15 amp fuse or circuit breaker. We recommend an individual branch circuit that serves only your washer.

WARNING: Never use an extension cord.

Electrical grounding is required for your washer.

You must ground your washer. In the event of a malfunction or breakdown, grounding reduces the risk of electric shock by providing a path of least resistance for the electric current.

Your washer is equipped with a power cord with a three-pronged grounding plug for use in a correctly installed and grounded outlet.

The outlet must be correctly installed and grounded in accordance with all local codes and ordinances.

WARNING:

- You are responsible for providing adequate electrical services for your washer.
- An incorrectly connected equipment-grounding conductor can increase the risk of electrical shock. Check with a qualified electrician or service person if you are not sure whether your washer is grounded correctly.
- Do not modify the power cord plug that came with your washer. If the plug does not fit the outlet, have the correct type outlet installed by a qualified electrician.
- To prevent unnecessary risk of fire, electrical shock, or personal injury, all wiring and grounding must be done in accordance with the National Electrical Code ANSI/FNPA, No.70 Latest Revision and local codes and ordinances.
- Never connect the ground wire to plastic plumbing lines, gas lines, or hot water pipes.
- The electrical valve, retractor, motor and drain pump of this appliance are intentionally not grounded and may present a risk of electric shock during serving. Do not contact these electronic devices while the appliance is plugged in.

Water requirements

To fill your washer in the correct amount of time, the water pressure needs to be between 14 and 116 psi (100 to 800 kPa).

If the water pressure is less than 14 psi (100 kPa):

- The water valve may fail or may not shut off completely.
- The time it takes to fill your washer may be longer than the time your washer controls allow. Your washer will turn off and report an error. A time-limit is built into the controls to prevent flooding in your home if a hose becomes loose.
- Water inlet hoses must be purchased separately. You can buy inlet hoses in various lengths up to 8 ft. (240 cm).

CAUTION: To avoid the possibility of water damage:

- Make sure that the water faucets are easily accessible.
- Turn off the faucets when you are not using your washer.
- Periodically check to make sure that water is not leaking from the water hose fittings.

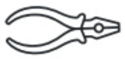
Drainage requirements

Note: The drain hose is attached to your washer at the factory.

- The height for the drain standpipe must be no shorter than 39 in. (99 cm) and no higher than 96 in. (244 cm).
- The drain hose must be routed through the drain hose clip to the standpipe.
- The standpipe must be large enough to accept the outside diameter of the drain hose.

03 INSTALLATION

Tools you will need


Pliers


Level


Adjustable wrench
(opens to 1 in. [2.5 cm])

Step-by-Step instructions

Step 1: Select a location

Before you install your washer, make sure that the location:

- Is a hard, level surface without carpeting or soft flooring that can obstruct ventilation.
- Is away from direct sunlight.
- Has adequate ventilation.
- Is not exposed to freezing temperatures (32° F or 0° C or below).
- Is away from heat sources, such as oil or gas.
- Has enough space so that your washer does not rest on its power cord.

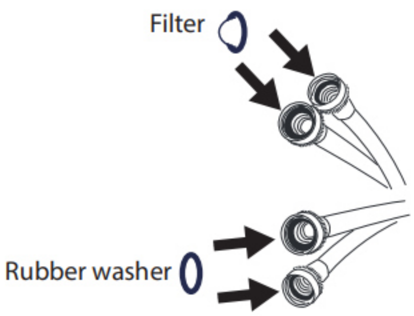
Step 2: Connect the water hoses

Note: The water supply hoses are not supplied with your washer and must be purchased separately.

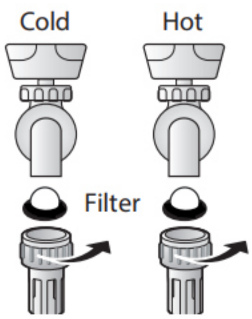
CAUTION:

- Use new water supply hoses. Using old hoses can result in leakage or overflow that can damage your property.
- Do not connect multiple water supply hoses together to increase the length of the hose. Hoses connected this way can leak and cause electrical shock. If a hose is too short, replace the hose with a longer, high-pressure hose.

1. Make sure that there are rubber washers inside the fittings on one end of each new water hose. A filter with an integrated washer will be installed on the other end of each hose. Installing the water supply hoses without rubber washers can cause water leakage.



2. Insert a filter (provided) in the input end of both hoses, then tighten the hose and faucet fittings by hand until they are snug. Tighten them an additional two-thirds of a turn with pliers or a wrench. Pull the water supply hoses downwards to make sure that they are connected securely.



CAUTION: Do not overtighten the fittings or apply tape or sealant to the faucets or water supply intakes. This can damage the fittings.

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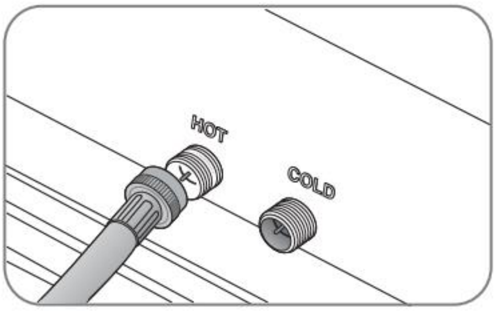
3 Place the unconnected ends of the hoses into a bucket and turn on the faucets connected to the water supply hoses for 10 or 15 seconds to remove any foreign substances. Turn off the faucets.

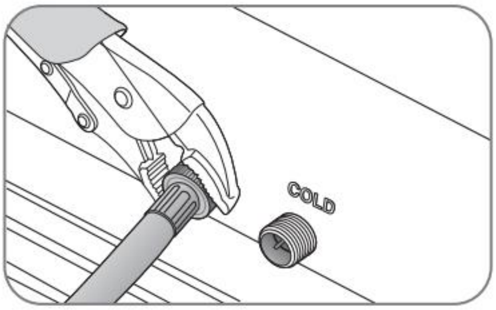
4 Connect the ends of the water supply hoses to the water supply intake connections at the top of your washer. Make sure that the rubber washers are in place. The water supply hose connected to the hot faucet must be connected to the hot water supply intake and the water hose connected to the cold faucet must be connected to the cold water supply intake.



Notes:

- If you do not want to use the hot water supply hose, insert a water intake cap into the hot water supply intake hole. In this case, you must select the **Tap Cold** option for the water temperature.
- No intake cap is provided with your washer. It must be purchased separately.

5 Tighten the fittings by hand until they are snug, then tighten them an additional two-thirds of a turn with a wrench.



CAUTION:

- Do not overtighten the fittings or apply tape or sealant to the faucets or water supply intakes. This can damage the fittings.
- Make sure that the water supply hoses are not twisted or bent. A bent or twisted hose can leak and cause an electric shock due to the water leakage. To ensure the correct water usage, connect both hot and cold water faucets. If either or both are not connected, an error code could occur.

6 Turn on the hot and cold water supplies and check all the water supply intake connections and the faucets for water leaks.

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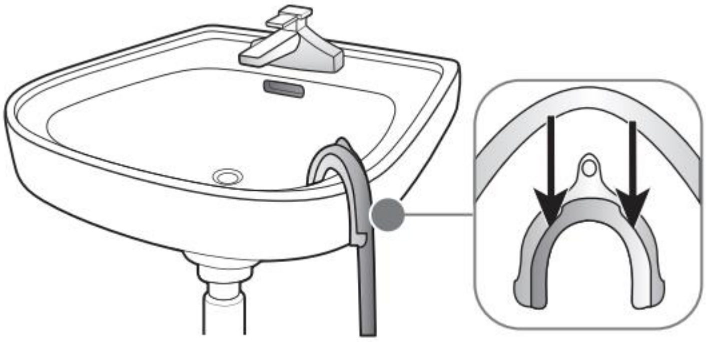
Step 3: Connect the drain hose

CAUTION:

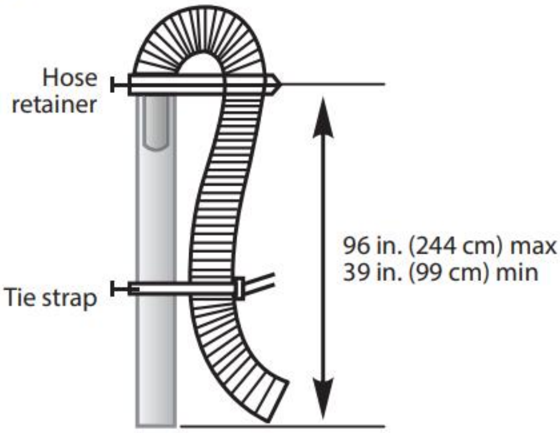
- Make sure that the connection between the drain hose and the wash basin, standpipe. or laundry tub is not airtight.
- Failure to slide the drain hose completely over your washer's drain pipe may result in water leakage.
- Failure to move the clamp or hoop onto your washer's drain pipe will result in water leakage.

Position the other end of the drain hose (the loose end) in one of the following ways:

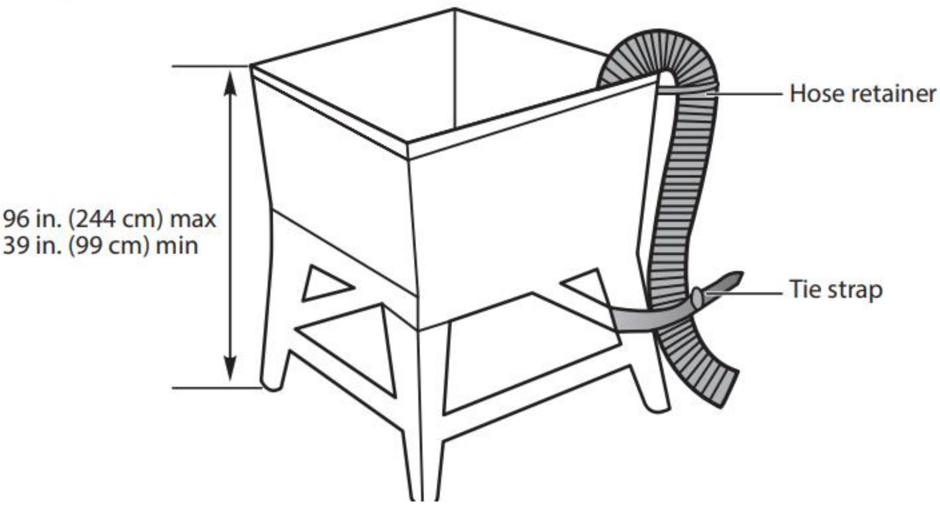
- **Over the edge of a wash basin** - Secure the guide to the side of the basin wall with a hook, tape it, or tie it with a piece of cord to prevent the drain hose from moving.



- **In a standpipe** - The standpipe must be no shorter than 39 in. (99 cm) and no higher than 96 in. (244 cm).



- **In a laundry tub** - The laundry tub must be no shorter than 39 in. (99 cm) and no higher than 96 in. (244 cm) maximum.



03 INSTALLATION

Step 4: Level your washer

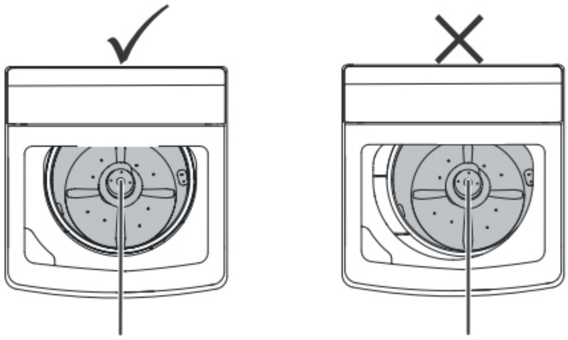
CAUTION:
Do not use your washing machine without leveling. If your washer is not level, it may:
- Vibrate excessively which can cause your washer to malfunction
- Make excessive noise.
- Indicate error codes E3 or E4 when your washer is running.

Extend the leveling feet only as much as is necessary. If the feet are extended too much, your washer may vibrate.

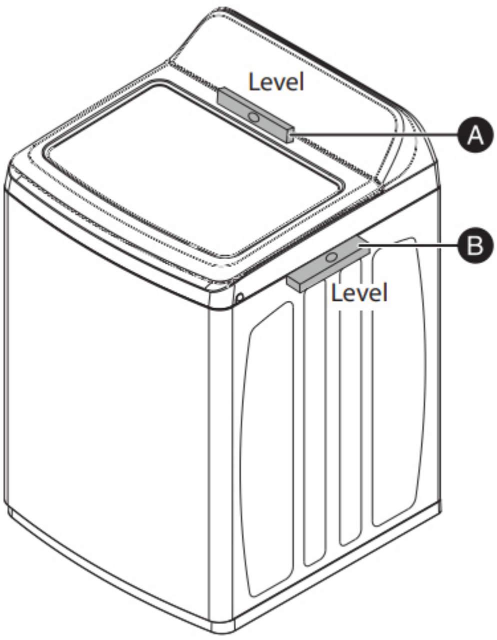
Determine if your washer is level by checking the position of the tub or by using a level.

1 Slide your washer into position.

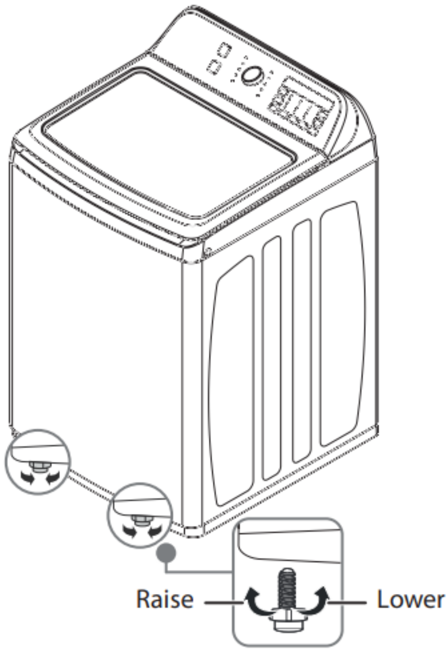
2 Open the lid and look from above to see if the tub is centered. If the tub is not centered, readjust the front feet. If you have a level, check the washer with the level.



Note: To check if the washer is level from side to side, place a level on the top of the washer, behind the lid (see 'A' in the image below). To check if the washer is level from front to back, slide a sheet of thin, stiff cardboard (such as the back of a legal pad) into the seam between the top and side panels of the washer on the right and left sides, then rest the level on top of the cardboard (see 'B' in the image below). Do not place the level on top of the washer.

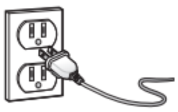


3 If your washer is not level, carefully tilt your washer just enough to adjust the leveling feet on the bottom front of your washer. Adjust the leveling feet. Turn all four leveling feet in one direction to raise the washer or the other direction to lower it.



4 Recheck to make sure that your washer is level. Push or rock the top edges of your washer gently to make sure the washer does not rock. If the washer rocks, readjust the leveling feet. If the lid does not stay open properly, extend the front leveling feet until the washer is level from front to back.

Step 5: Power your washer



Plug the power cord into a well-grounded, 3-prong, 120V AC at 60Hz approved electrical outlet, protected by a 15 amp fuse or comparable circuit breaker. Your washer is grounded through the third prong of the power cord.

Step 6: Run a test cycle

Run a test cycle to make sure that your washer is properly installed.

1 Load about 6 pounds (2.7 kg) of laundry into the tub.

2 Press the ⏻ **Power** button to turn on your washer.

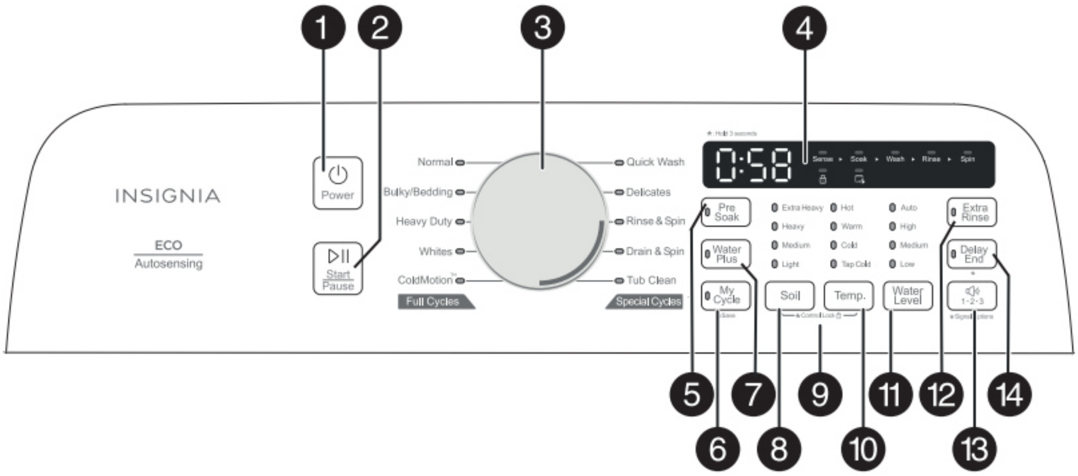
3 Turn the cycle selector to select the **Rinse & Spin** cycle.

4 Press the ▶II **Start/Pause** button to start the test cycle. Your washer should not rock or vibrate excessively when water fills the tub, or when washing or spinning. Your washer should drain well during the spin cycle.

CAUTION:
- If you detect any water leaks while your washer is filling or draining, check the water connections. See Step 2: Connect the water hoses on page 12 or Step 3: Connect the drain hose on page 14.
- If your washer rocks and/or vibrates excessively, level it again. See Step 4: Level your washer on page 15.

04 OPERATION INSTRUCTION

Control panel



#	ITEM	DESCRIPTION
1	⏻ **Power** button	Press to turn on your washer. Press again to turn off your washer. If you leave your washer on and do not press the ▶II **Start/Pause** button within 10 minutes, your washer automatically turns off.
2	▶II **Start/Pause** button	Press once to start the wash cycle. Press again to pause the cycle. If you pause the cycle, you can add or remove items, but you cannot change any settings. Press again to restart the cycle.
3	Cycle selector knob	Turn to select a wash cycle. The cycle you select determines the wash pattern for the cycle. For more information, see Washer options and settings on page 20.
4	Digital display	• The initial display is the total time for the cycle you select. • While your washer is running, the display shows the cycle process and time remaining in the cycle. **Note**: The time shown is an estimate based on normal operating conditions. External factors (such as the load size, room temperature, incoming water temperature, and water pressure) can affect actual time. • If you set a delay time for the start of the cycle, the display shows the time when the cycle will end. • Your washer has a power-off memory feature. When the power is abnormally cut off to your washer and is later restored, it will continue the current cycle. It displays the total time of the cycle first, then, after four seconds, it will display the remaining time of the cycle.
5	**Pre Soak** button	Press to select the **Pre Soak** option. Press again to cancel. Select this option for heavily soiled items that need to soak to remove stains and heavy dirt. Pre Soak occurs after the tub fills and detergent is added, but before the wash process. The agitator will rotate several times during the process.
6	**My Cycle** button	Press and hold for three seconds to save your favorite washing cycle and options. When you use it next, press once to load it.
7	**Water Plus** button	Press once to increase to the next higher water level. Press again to cancel this function. If the water level is already set to the highest, the level will not change.
8	**Soil** button	Press to cycle through the available soil level settings. Selections include **Extra Heavy**>**Heavy**>**Medium** >**Light** (indicated by the soil level LEDs). Different soil levels result in different washing times and wash cycle settings. For more information, see Washer options and settings on page 20.
9	**Control** (child) **lock**	Press and hold the **Soil** and **Temp.** buttons at the same time for three seconds to turn on the child lock function. Press and hold these buttons again for three seconds to turn off the function. When the child lock function is turned on, the only button that works is the ⏻ **Power** button and the **Soil** and **Temp.** buttons you use to turn off the child lock.
10	**Temp.** button	Press to cycle through the available washing water temperatures. Selections include **Hot**>**Warm**>**Cold**>**Tap Cold** (indicated by the water temperature LEDs). The rinse water temperature is always cold.
11	**Water Level** button	Press to cycle through the available water level settings. Selections include **Auto**>**High**>**Medium**>**Low** (indicated by the water level LEDs). Different water levels result in different washing times and wash cycle settings. For more information, see Washer options and settings on page 20.
12	**Extra Rinse** button	Press to add an extra rinse to the wash cycle. Press again to cancel.

04 OPERATION INSTRUCTION

#	ITEM	DESCRIPTION
13	1·2·3 button	Press and hold the button for three seconds to choose a sound notification mode. Three different sound modes can be selected: Mode 1: All sounds (sound of button and cycle end) are on. Mode 2: Sound of button is off, the sound of cycle end is on. Mode 3: All sounds are off.
14	**Delay End** button	Press and hold this button for three seconds to initiate this function. The initial setting delays the end of the wash cycle by one hour. Each successive button press adds an additional hour (up to 12 hours). Once the setting is complete, press the ▶II **Start/Pause** button (if no action is taken, the function will automatically cancel after five seconds). The wash cycle will start at the appropriate time in order to finish at the time you set. To cancel the end-of-wash cycle delay, press this button multiple times until zero is displayed, or press and hold this button to quickly advance though the settings back to zero. If the cycle you have selected takes longer than one hour, the number 1 will not be displayed because the end time is not possible to delay. For example: **Normal** Cycle will jump from 0 to 2 because it takes an hour to run the cycle and therefore cannot be set to 1. A modified cycle like **Quick Wash** could be set to 1.

Cycle	Fabric type	Soil level	Temp.	Water level	Pre Soak	Water Plus	My Cycle	Extra Rinse	Delay End	Signal* 1*2*3
Normal	Cotton, linen Towels, Shirts Sheets, Jeans Mixed loads	Extra Heavy Heavy **Medium** Light	Hot **Warm** Cold Tap Cold	**Auto** High Med Low	●		●	●	●	●
Bulky/ Bedding	Large items like blankets and comforters	Extra Heavy Heavy **Medium** Light	Hot **Warm** Cold Tap Cold	**Auto** High Med Low	●	●	●	●	●	●
Heavy Duty	Heavily soiled cotton fabrics	**Extra Heavy** Heavy Medium Light	**Hot** Warm Cold Tap Cold	**Auto** High Med Low	●	●	●	●	●	●
Whites	White Fabric	Extra Heavy Heavy **Medium** Light	Hot **Warm** Cold Tap Cold	**Auto** High Med Low	●	●	●	●	●	●
Cold Motion	Cotton, linen Towels, Shirts Sheets, Jeans Mixed loads	Extra Heavy Heavy **Medium** Light	**Tap Cold**	**Auto** High Med Low	●	●	●	●	●	●
Quick Wash	Lightly soiled and small loads	Medium **Light**	Hot **Warm** Cold Tap Cold	Med **Low**	●	●	●		●	●
Delicates	Dress shirts Blouses Nylons Sheer/lacy	Heavy Medium **Light**	Hot Warm **Cold** Tap Cold	**Med** Low	●	●	●	●	●	●
Rinse & Spin			**Tap Cold**	**Auto** High Med Low			●		●	●
Drain & Spin							●			●
Tub Clean		**Light**	**Hot**	**High**			●			●

* Table in grey is an initial setting. Dots are all optional functions you can select.

05 TEST MODE

Check Method	Step	Check Item
Within 10 seconds of turning on or off the power, press **My Cycle** button and **Soil** button together then Press **Power** button to enter test mode. Anytime you enter the test mode, the digital will shows '**FCt**', If the digital in not '**FCt**' displayed, replace the PCB.	Enter the Test mode	Enter the Test mode
When the digital shows '**FCt**', press **My cycle** button to enter Washing test. In 3-6 seconds, the motor and impeller will rotate in two-directions in turn. The digital will shows '**L1**'. The test mode will end after 3 minutes' operation or press button **Start/Pause** back to '**FCt**'.	Test 1	**Washing test,** to check the motor and rotate directions.
When the digital shows '**FCt**', press **Temp.** button to enter Spining test. a) After 3 seconds, the clutch and pump will power on. Then the drum will rotate forward in 7s and the digital will shows the number (variable frequency machine) / sp (normal machine). The test mode will end after 3 minutes' operation or press button **Start/Pause** back to '**FCt**'. b) If the lid is open, the digital will shows '**E3**' error. The error will be solution when the lid is closed. '**E4**' error solution: Open the lid cover.	Test 2	**Spinning test,** to check clutch and pump.

05 TEST MODE

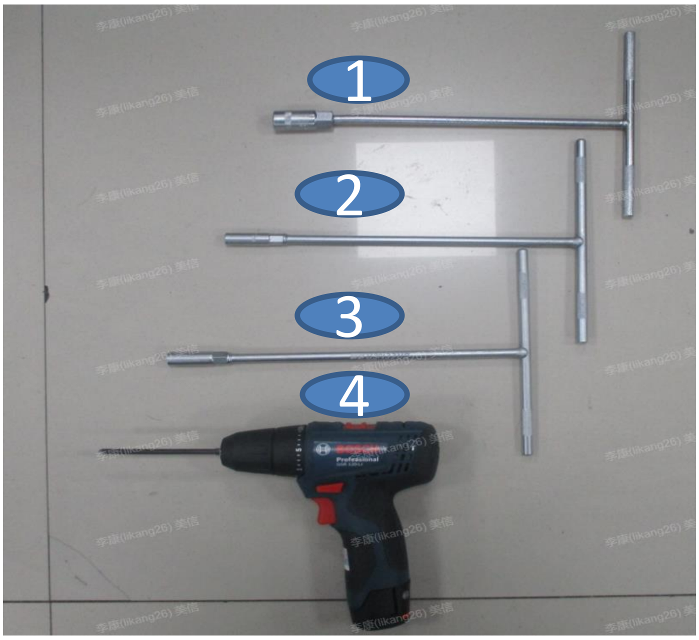
Check Method	Step	Check Item
When the digital shows '**FCt**', press **Water Level** button to enter Fuzzy Weighing test. The unit will check the auto-sensing function in this test mode. In this check, the motor will rotate in a cycle at a certain beat after 1-6s. After the motor stops, the digital will shows the actual pulse number. The test mode will end after 1 minutes' operation or press button **Start/Pause** back to '**FCt**'.	Test 3	**Fuzzy Weighing check**
When the digital shows '**FCt**', press **Soil** button to enter Inlet valve and Software version check. The valves will be ON and water can enter the unit. Check the detergent compartment, softener compartment and the basket water filling to check every valve. At the same time, the digital will shows in sequence: "Capacity", "Project number", "Display PCB software version number", "Driver PCB version number", "Lower limit of bucket collision parameters", "Upper limit of bucket collision parameters", "temperature (if the machine has intelligent temperature sensing). The test mode will end after 1 minutes' operation or press button **Start/Pause** back to '**FCt**'. The version number information is in the software version management table. For example, Capacity is "C16", Project number is "v10", Display PCB software version number is "v100", Driver PCB version number is "08", Lower limit of bucket collision parameters is "02", Upper limit of bucket collision parameters is "30"	Test 4	**Inlet valve and Software version check**

05 TEST MODE

Check Method	Step	Check Item
After pressing the Power button, press button **Mycycle**, **Signal**, **Mycycle**, **Signal** successively within 10s enter Continuous mode. In this mode, the program light is all on, and the corresponding light of the selected will be blink. After setting the parameters, press the **Start/Pause** button and the machine will run in accordance with the set program parameters. This mode will end when power off.	Test 5	**Continuous mode**
When the digital shows ‘**FCt**’, turn the **Cycle Selector** to ‘**t01**’, then press the **Start/Pause** button enter the Internal version number querying. The digital will shows "Internal display PCB software version number". This mode will end when press button **Start/Pause** back to ‘**FCt**’.	Test 6	**The Internal version number querying**

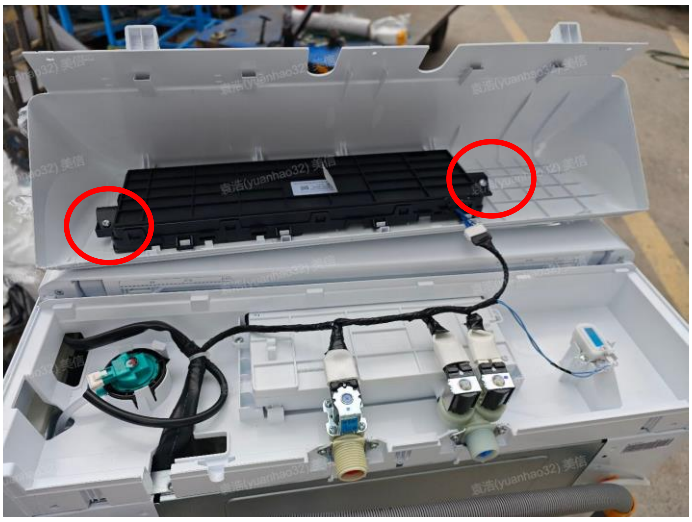
06 Disassembly of Washing machine

Service tools needed

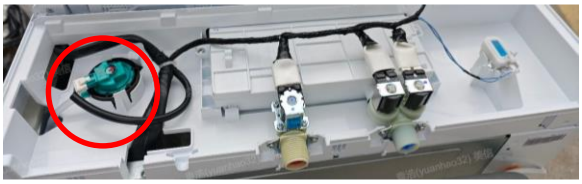
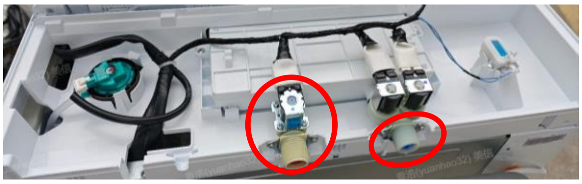
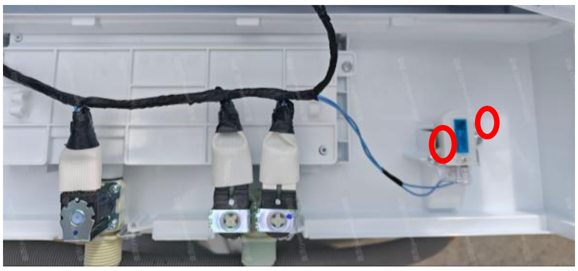


Number	Tools
1	13# sleeve spanner
2	10# sleeve spanner
3	8# sleeve spanner
4	6# inner six angle wrench

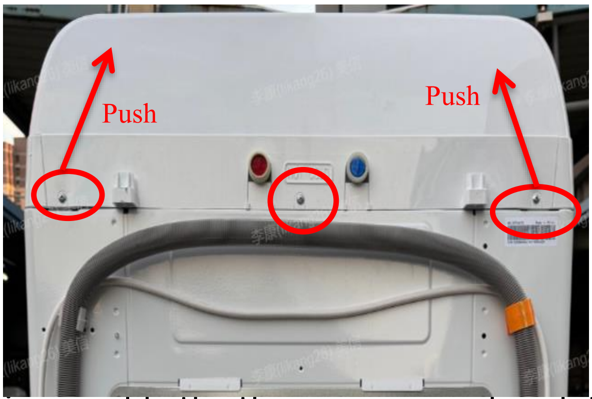
06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
Check and Replace the PCB(LV) Water inlet valves, Water Level Switch, Impact switch	1.1 Remove the Control Panel Assembly	 Disassemble the three screws fixing the control panel Assembly on the back of the unit. Push the control panel front to open.	Screwdriver
	1.2 Remove the Knob Assembly	 Pull the knob out.	No tools needed
	1.3 Disassembl e the PCB(LV)	 Disassemble the screws and Housing.	Screwdriver

06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
	1.4 Disassembly and check water level sensor	 	Screwdriver
Check and Replace the PCB(LV) Water inlet valves, Water Level Switch, Impact switch	1.5 Disassembly and check the water inlet valve	 	No tools needed
	1.6 Disassembly and check the Impact switch	 	Screwdriver

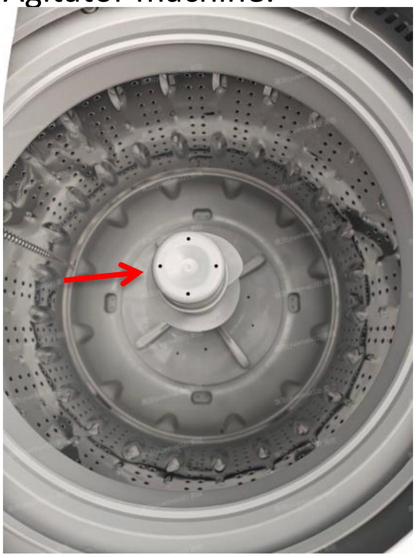
06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
Check and Replace the Door lock	2.1 Remove the Top Cover Assembly	 Disassemble the three screws and push top cover from back side	Screwdriver
	2.2 Open the top cover assembly	 Open the top cover assembly sidelong but be careful for impact switch no brake.	No tools needed
	2.3 Open the Lock box	 Disassemble the two screws and open the lock box	Screwdriver

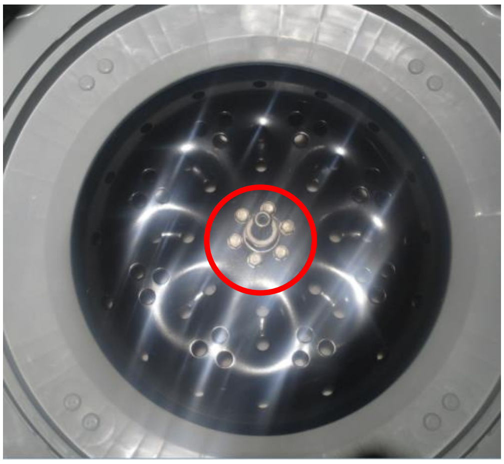
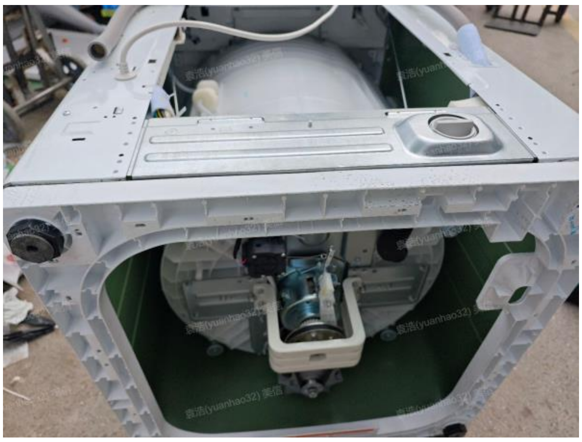
06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
Check and Replace the Door lock	2.4 Disassembly and check the Door lock		Screwdriver
Check and Replace the Electrical PCB	3.1 Remove the Rear cover	 Disassemble the four rear cover screws and four Electrical Box screws.	Screwdriver
	3.2 Remove the Electrical Box	 Take the Electrical Box out of cabinet	No tools needed

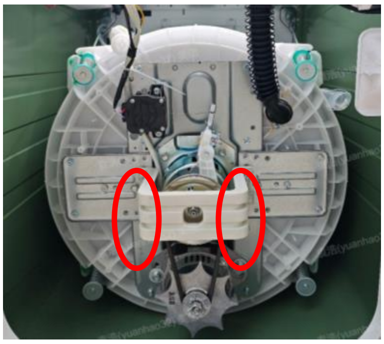
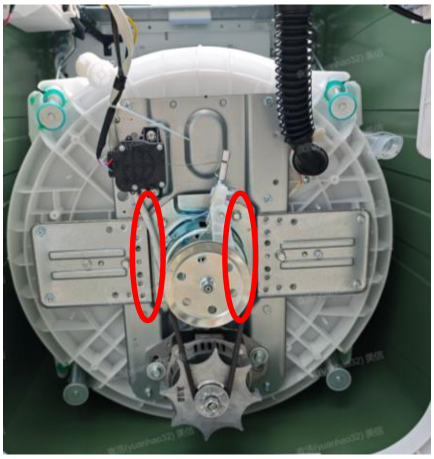
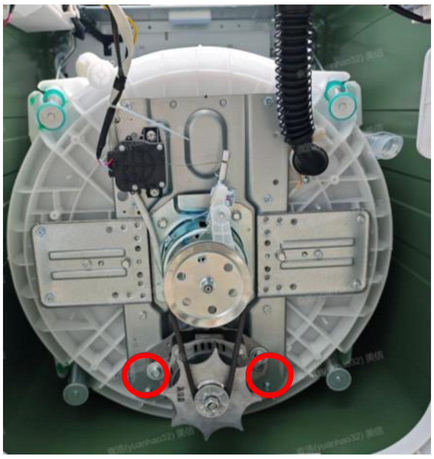
06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
Check and Replace the Electrical PCB	3.3 Open the Electrical Box	 Open the electrical box.	Screwdriver
	3.4 Check the Electrical PCB		No tools needed
Check and Replace the Clutch , Motor, Drain pump, Retractor	4.1 Remove the Muddler Cover	Agitator machine: 	No tools needed

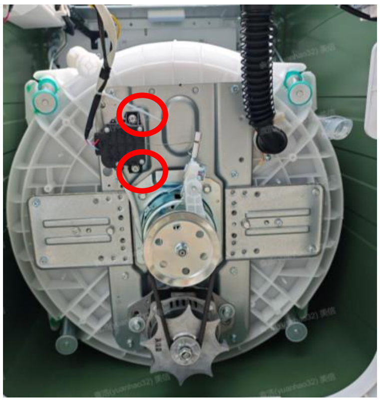
06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
Check and Replace the Clutch , Motor, Drain pump, Retractor	4.2 Remove impeller/ Agitator screw	Agitator machine: Disassemble the one screw and take Agitator out. 	Screwdriver
	4.3 Disassemble six screw of clutch	 Disassemble six screw of clutch	No.08 Sleeve
	4.4 Place down of laundry		No tools needed

06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
Check and Replace the Clutch , Motor, Drain pump, Retractor	4.5 Disassemble the bracket	 Disassemble four screw of bracket by sleeve	Ten sleeve
	4.6 Disassemble and check the clutch body subassembly		No.10 Sleeve
	4.7 Disassemble and check the Motor		No.13 Sleeve
	4.8 Disassemble and check the Drain Pump		No.10 Sleeve

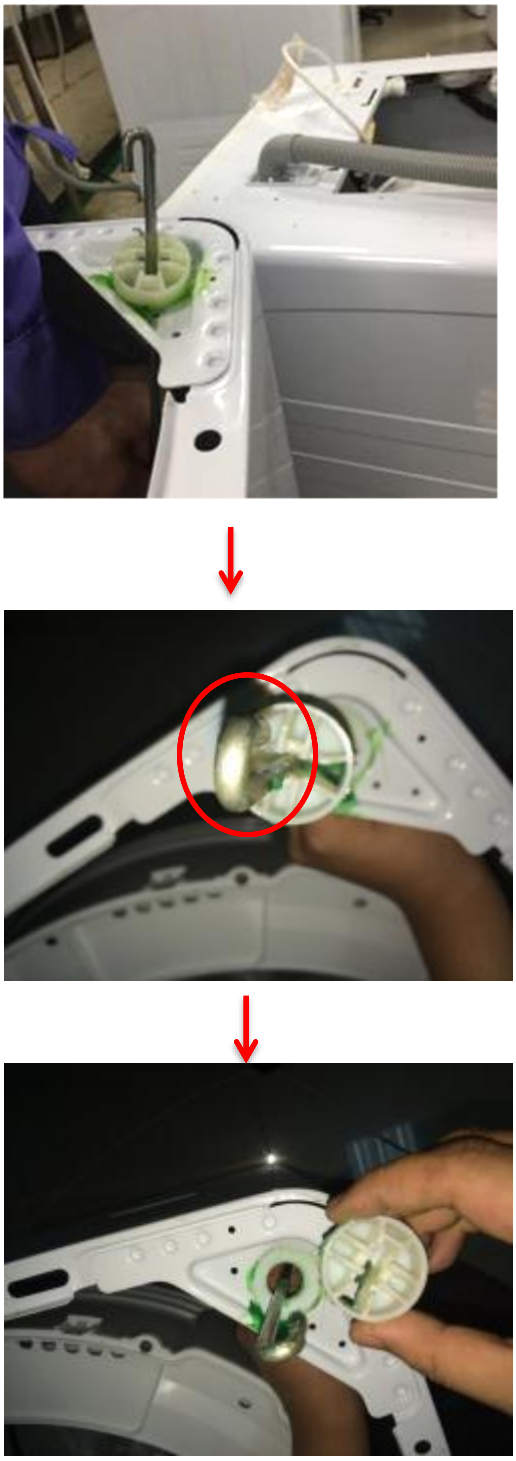
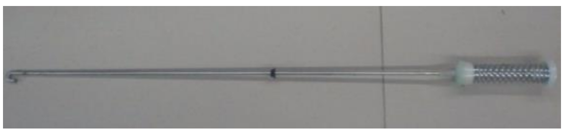
06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
Check and Replace the Clutch , Motor, Drain pump, Retractor	4.9 Disassemble and check the Retractor		No.10 Sleeve
Check and Replace the suspend system	5.1 Remove the Top Cover Assembly	Same as 2.1	Screwdriver
	5.2 Open the top cover assembly	Same as 2.2	No tools needed

06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
Check and Replace the suspend system	5.3 Remove the suspend out with Outer Tub	  	No tools needed

06 Disassembly of Washing machine

What's for	Steps	Description	Tools need
Check and Replace the suspend system	5.3 Remove the suspend out with Outer Tub		No tools needed
	5.4 Remove the suspend out with Cabinet.		No tools needed

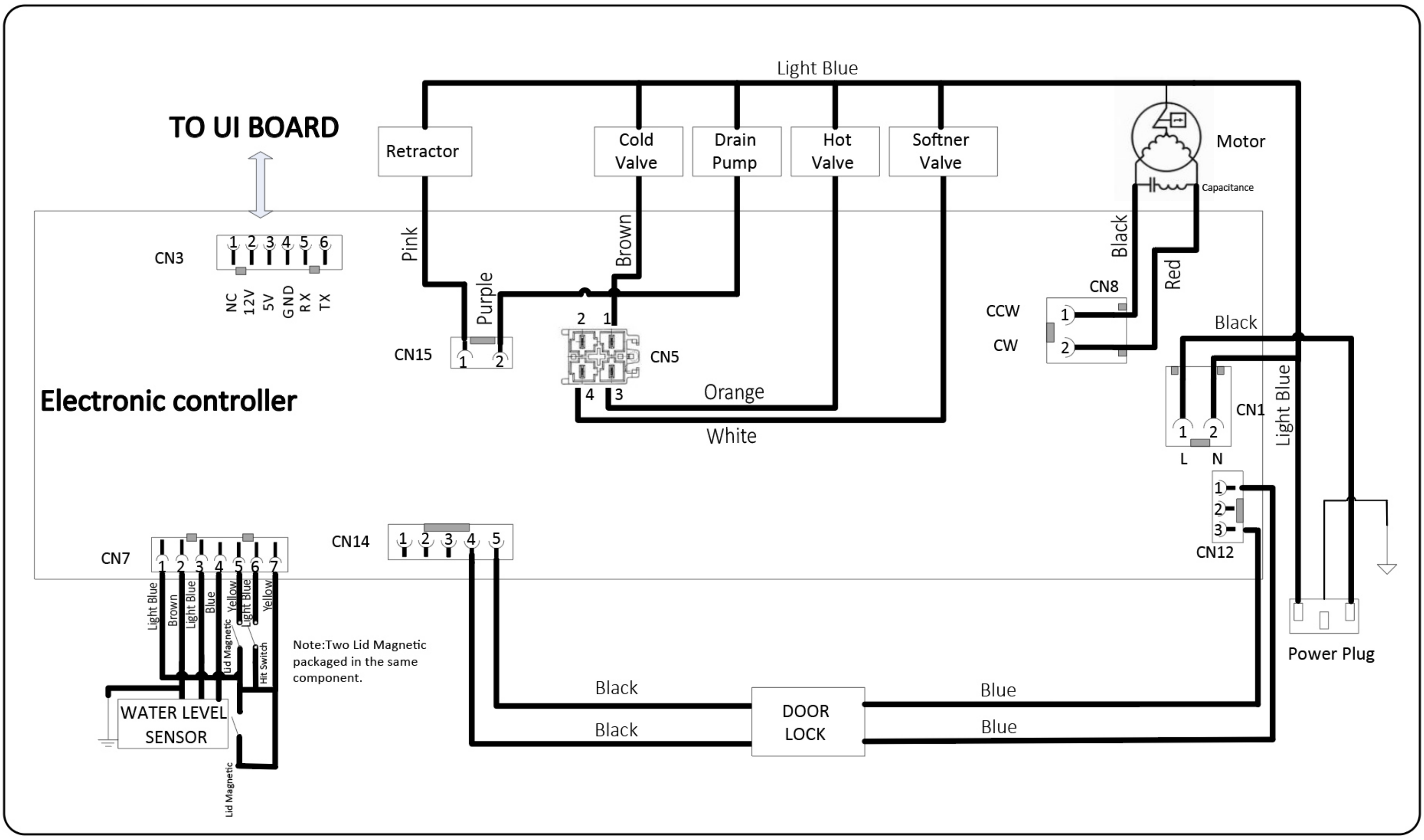
07

ERROR CODE TROUBLESHOOTING

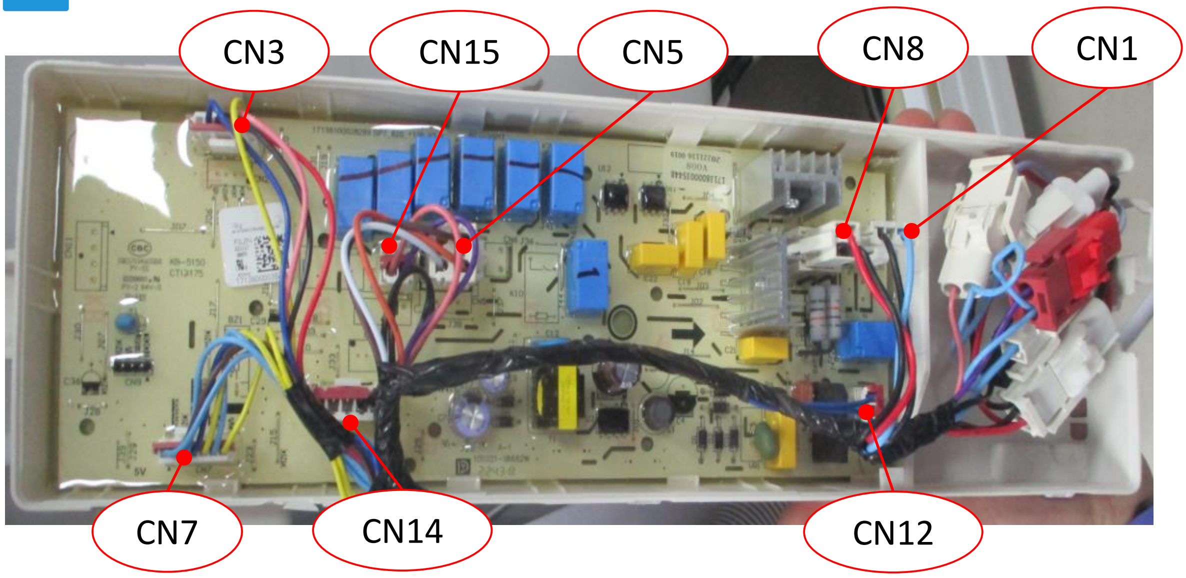
TOP LOADING

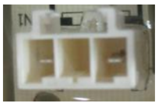
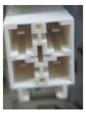
07 TROUBLESHOOTING

Circuit diagram



07 TROUBLESHOOTING



Port	Name on PCB	Definition
CN1		Pin 1: Live line Pin 3: Null line
CN3		PIN1: TXD PIN4: +5V PIN2: RXD PIN5: +12V PIN3: GND PIN6: NC
CN4		PIN1: NC PIN2: NC PIN3: NC PIN4: Door Lock status detection PIN5: Door Lock status detection
CN5		PIN1： Cold valve PIN2： Detergent valve PIN3： Hot valve PIN4： Softener valve
CN15		PIN1： Retractor PIN2： Drain pump
CN7		PIN1: Common PIN5: Reed switch PIN2: GND PIN6: Out of Balance Switch PIN3: Water level sensor PIN7: 12V PIN4: Water level sensor
CN8		PIN1: Negative rotation control PIN2: Positive rotation control
CN12		PIN1: Door lock control PIN2:NC PIN3: Door lock control

07 TROUBLESHOOTING

E1 Abnormal Water Inlet

Define: The water level fails to reach the set level after exceeding the set time, or the water level changes little within a certain period of time
Reasons: When water in the tub didn`t achieve the specified level in 30min,PCB will display “E1”.Remind the timeout. Some machine is 30min or more than 30min

Malfunction code	Unit Action Description	Possible cause
E1	There is no water or water is very small	The faucet is not turned on
		The inlet pipe is bent
		Low water pressure or water outage
		The water inlet valve is blocked, water inflow is very small
		The wiring of the inlet valve is disconnected or broken
		Water inlet valve damage does not work
		PCB failure.
	Water leakage, water is entering and draining at the same time	Water level sensor pressure pipe is loose,or water in the pipe
		Water level sensor failure
		PCB failure.

07 TROUBLESHOOTING

E1 Abnormal Water Inlet

Check Procedure:

- Check if the faucet is turned on
  - No → Turn on the faucet
  - Yes ↓
- Check if the inlet pipe is bent
  - Yes → Adjust the bent part of the inlet pipe
  - NO ↓
- Check if the water pressure is too low or if there is a water outage
  - Yes → Contact the water supply company to check the water pressure
  - NO ↓
- Disassemble the inlet pipe, check if the pipe or inlet valve is blocked
  - Yes → Remove any blockages
  - No ↓
- Check if the inlet valve wiring harness open-circuited
  - Yes → Connect wires or replace wires
  - No ↓
- Whether the inlet valve is damaged
  - Yes → Replace the inlet valve
  - No ↓
- Water leakage, water is entering and draining at the same time
  - Yes → Adjust or replace
  - No ↓
- Check if the bucket can be filled with water
  - No → Adjust or replace the water level sensor
  - Yes ↓
- Check if the air pressure tube is leaking
  - Yes → Adjust or replace the water level sensor
  - No ↓

Replace the PCB check whether the function is OK

07 TROUBLESHOOTING

E1 Abnormal Water Inlet

Check Procedure:

Step 1 Check if the faucet is turned on.

If the user has not turned on the faucet, then it needs to be turned on (Fig.1)

Fig.1

Step 2 Check if the inlet pipe is bent

The inlet pipe should be smoothly connected. If there is any bending, adjust the bending point.(Fig.2)

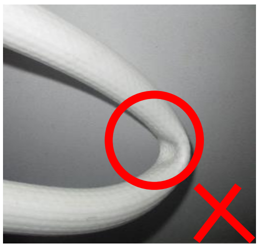
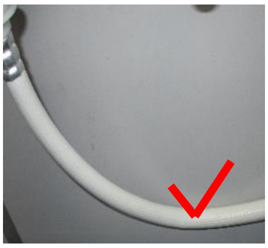


Fig.2



Step 3 Check if low water pressure or water outage

Turn on the faucet, unscrew the inlet pipe, and check if the water pressure is too low or if there is a water outage

Step 4 Check if Inlet valve blockage

Remove the inlet pipe, check if there is any foreign object blocking the inlet valve connection, clean it thoroughly, and reinstall it

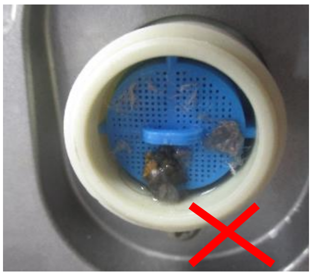
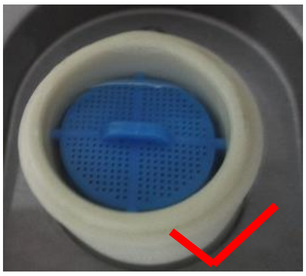
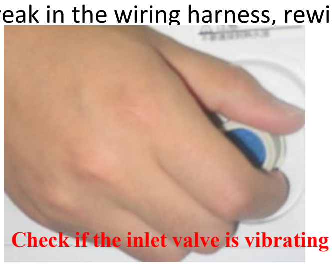


Fig.3



Step 5 Check if Inlet valve wiring circuit break

1. Adjust the program to the water inlet state, touch the inlet valve with your hand to check if there is a vibration and a weak electromagnetic hum.
2.If there is no such phenomenon, remove the workbench and use a multimeter to check if there is voltage at the two terminals of the valve.
3. If there is AC voltage, refer to step 6) and directly replace the inlet valve. If there is no AC voltage, check if there is a circuit break in the wiring harness.
4. For non-frequency conversion machines, professionals can refer to the following steps ① and ② to check if there is a circuit break in the wiring harness, rewire it and do a good job of insulation protection, or replace the wire.



07 TROUBLESHOOTING

E1 Abnormal Water Inlet

①Close the power supply and unplug the power cord. Remove the computer board and locate the terminals marked pin “1” of “CN5” on the board. ((Fig.1, Fig.2). These are the “L” poles that supply power to the inlet valve. Check if the terminals are loose. And find the light blue wire bundle(Fig.3). This is the "N" pole that supplies power to the inlet valve.

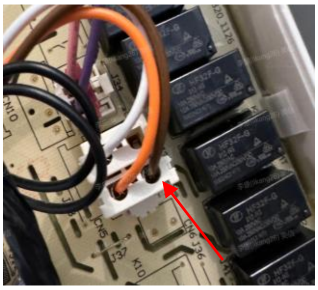


Fig.1


Fig.2


Fig.3

②Connect one end of the multimeter to the terminal connected to the wire bundle inserted in this position, and the other end of the multimeter to the terminal of the inlet valve inserted in the wire bundle. Set the multimeter to the continuity test mode. If the wire bundle is a circuit, the multimeter will show a normal reading. (Note: the same wire color should be measured separately.)


Fig.4

③If the wiring harness is not open-circuited, it is necessary to check whether the computer board is normal. In this case, the washing machine needs to be powered on and adjusted to the water inlet state (do not unplug any terminals). Use a multimeter to connect one end to the port labeled "JS" on the computer board and the other end to the port labeled "N" to check whether the voltage is the same as the mains voltage. If there is no mains voltage, proceed to step 8 and replace the computer board. (The following diagram shows two typical computer boards, and the port locations may vary for different computer boards. For frequency conversion models, it is recommended to directly replace the inverter.)

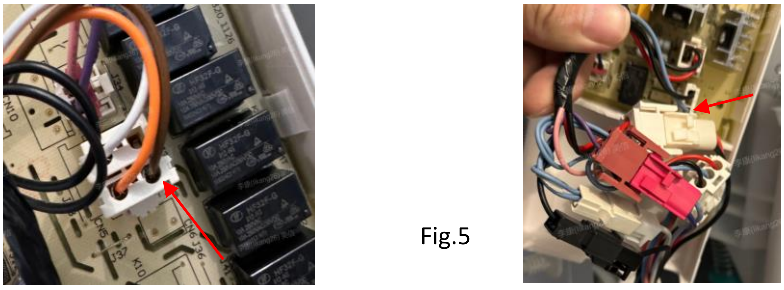


Fig.5

07 TROUBLESHOOTING

E1 Abnormal Water Inlet

Check Procedure:

Step 6 Check if Inlet valve wiring circuit break.

If the inlet valve is damaged, and in step 5) it is found that there is mains voltage at both ends of the inlet valve but the inlet valve does not work, then replace the inlet valve.

Step 7 Check if the washing machine is leaking.

If there is water leakage and water is being simultaneously filled and drained, and the inlet valve is always on but the tub cannot hold water.

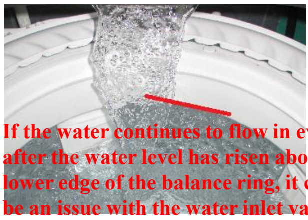
① Check if there is any water leakage at the water tank and the clutch installation. First, check if there are any damaged or leaking points on the water tank, and if there is any water leakage at the clutch installation hole, replace it if it is damaged.

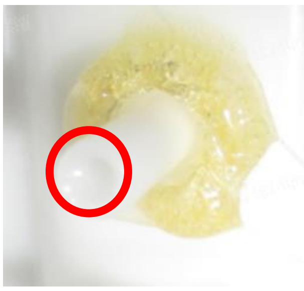
②Check if the Drain Hose is hung up beyond the height of 100cm. The drain hose end should be 39~96 in. height from the bottom of the unit.

Step 8 Check the water level sensor

Water level sensor malfunction. If the inlet valve keeps filling with water and the water level is close to or above the balance ring, check if the air return pipe is leaking, blocked, or bent, or if the water level sensor is damaged.

① If the issue is caused by a leaking or bent air return pipe or a blocked air return chamber, it needs to be adjusted. If the air return pipe is normal, the water level sensor needs to be replaced.





Step 9 Check if the computer board is damaged

If so, replace the computer board or inverter. Professional personnel can refer to Step 5- ③ for testing procedures.

07 TROUBLESHOOTING

E2 Drainage Problems

Define: The drainage time has exceeded the set time and has failed to drain completely.
Reasons: When water in the tub didn`t drain out absolutely in 10min,PCB will display “E2”.Remind the timeout. Some machine is 15min or more than 15min

Malfunction code	Unit Action Description	Possible cause
E2	The water in the tub didn’t drain out absolutely	Drainage pipe is squeezed or bent.
		Drain-pipe/ Inner drain-pipe/ Drain valve / Drain pump filter is blocked
		The terminals of the traction motor poorly connected or the wiring harness broken
		Drain pump damage
		PCB failure.
	The water in the tub is drain out absolutely	Water level sensor failure
		PCB failure.

Check Procedure:

Check if the drainage pipe is crushed or bent → Yes → Straighten the drainage pipe

No ↓

Check if Drain-pipe/ Inner drain-pipe/ Drain valve / Drain pump filter is blocked → Yes → Remove the blockage

No ↓

Check if the terminals of the traction machine are loose or if the wiring harness is broken → Yes → Adjust or replace the traction wire rope or rod

No ↓

Check if the drain pump is damaged → Yes → Replace the Pump

No ↓

Replace the PCB check whether the function is OK

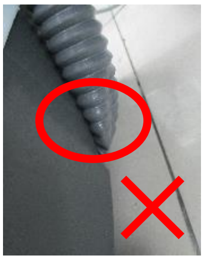
07 TROUBLESHOOTING

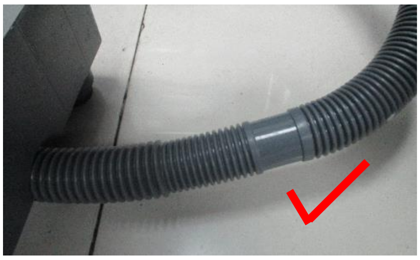
E2 Drainage Problems

Check Procedure:

Step 1 Check if the drainage pipe is crushed or bent.

A crushed or bent drainage pipe can cause poor drainage and trigger drainage timeout alarms, so it is necessary to ensure that the pipe is unobstructed.





Step 2 Check if Drain-pipe/ Inner drain-pipe/ Drain valve / Drain pump filter is blocked

If it is, manual cleaning is required.

补充Filter 堵塞照片

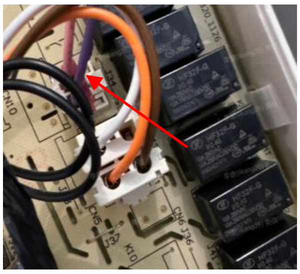
Step 3 Check whether the terminal of the pump motor is poorly connected or the wiring harness is broken.

If the pump motor does not work, check whether there is voltage at both ends of the pump motor. If the mains voltage is detected, replace the pump motor. Otherwise, it may be caused by poor terminal connection or broken wiring harness, which cannot supply power to the traction motor.

①First, check if there is any loose connection at the wire terminal, and adjust or replace the wiring if necessary；

②To check if there is a broken wire in the wire harness, you can directly inspect the appearance for any signs of damage or breakage. You can also use a multimeter to test the wire harness. The process is as follows：

a. Turn off the power, remove the computer board, locate the plug socket marked pin “1” of “CN15” on the computer board, connect one end of the multimeter to the terminal on the wire harness that is connected to “1” of “CN15”, and connect the other end to the output terminal of the wire harness that is connected to the "L" line (usually the red line) of the wire bundle and the traction machine.



b. Set the multimeter to the wire circuit continuity mode. If it is a complete circuit, the multimeter will display the correct indication.

07 TROUBLESHOOTING

E2 Drainage Problems

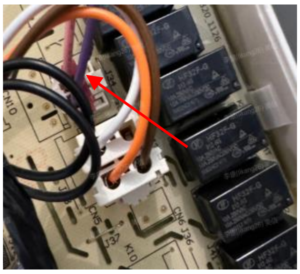
Check Procedure:

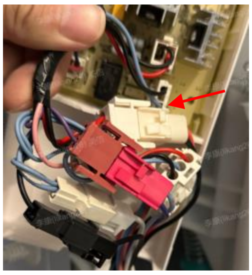
Step 4 If voltage is detected normally in step 3, it indicates that the pump is damaged and needs to be replaced, and then re-tested.

Step 5 Check if the computer board/inverter is damaged

If no voltage is detected on both ends of the Pump, and it is confirmed that the wiring harness is normal in step 3), then it is necessary to check whether the output terminals of the computer board are outputting voltage. If there is no output voltage, the computer board needs to be replaced.

①With the washing machine in the power-on state, switch to the spin mode, and use the multimeter to connect one end to the port marked pin “1” of “CN15” on the computer board and the other end to the port marked "N". Check if there is mains voltage present. If there is no voltage, then the computer board should be replaced.





Notes:

■Use stable and insulated methods when connecting the multimeter. Do not touch the multimeter connection to the human body or the washing machine shell.

■After the washing machine is powered on, personnel should not touch the multimeter or other washing machine circuits

■The voltage difference between the blue line and the red line is 5V, but the voltage difference between a human body and them is 120V. Please pay attention to electrical safety.

②For variable frequency washing machines, you can try to directly replace the frequency converter.

07 TROUBLESHOOTING

E3 Door Left Open

Define: During operation, the door is opened or the drum switch is disconnected.
Reasons: When the washing machine in the "spin" program, safety switch is not closed, cause the machine alarm.

Malfunction code	Unit Action Description	Possible cause
E3	Cover is opened or closed	The washing machine door is not closed
		The door magnetic switch and magnet undesirable
		The impact switch undesirable
		The wiring harness is open circuit
		PCB failure

Check Procedure:

Check if the washing machine door is closed. — No → Close the door and press the start button again (for some models, closing the door is enough).

↓ Yes

Check if the door magnetic switch and magnet are functioning properly — No → Replace the door magnetic switch or magnet

↓ Yes

Check impact switch — No → Replace impact switch

↓ Yes

Check whether the wiring harness is broken or not — Yes → Connect the wires properly or replace the wiring harness

↓ No

Replace the PCB check whether the function is OK

07 TROUBLESHOOTING

E3 Door Left Open

Check Procedure:

Step 1 Check if the user have closed the door.

During operation, if the door is not properly closed, an E3 alarm will occur. Advise the user to close the door properly and press the start button again. (In some models, simply closing the door will clear the alarm.)

Step 2 Check the door magnetic switch and magnet.

Open the workbench and try replacing the door magnetic switch. Move a magnetic-iron close to the panel as figure right to check if the Error Code disappeared. (it is located close to the door magnetic switch). A normal magnet will attract the magnetic-iron. After completing the above steps, test the machine again. If it functions properly, it indicates that the door magnetic switch is damaged. Professionals can use a multimeter to check the resistance of the door magnetic switch to determine if it is faulty.

With the door closed, check the resistance value of the door switch using a multimeter.

With the door open, check the resistance value of the door switch using a multimeter.

Step 3 If all the above steps are normal, then the next step is to replace the PCB

07 TROUBLESHOOTING

E4 Imbalance Error

Define: During the dehydration process, if the drum is in an unbalanced state and repeatedly hits the drum switch, it can prevent the dehydration process from functioning properly.
Reasons: Ouer tub hit the safety switch or cabinet seriously,the machine can`t spin smoothly

Malfunction code	Unit Action Description	Possible cause
E4	Machine is unbalanced	The washing machine shaking or leveled
		Clothes is unbalanced
		The impact switch failure

Check Procedure:

Check if the washing machine is shaking or level → Yes → Adjust level

↓ No

Check whether the clothes is placed evenly → No → Reload the clothes, and guide the user in selecting the appropriate water level and program.

↓ Yes

Machine is balanced,but it still shaking seriously

↓ Yes

Replace the impact switch

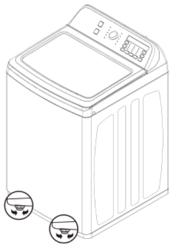
07 TROUBLESHOOTING

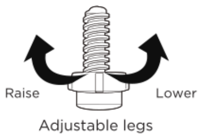
E4 Imbalance Error

Check Procedure:

Step 1 Check if the washing machine is shaking or level.

If the washing machine is tilted or shaking, it can trigger the drum switch when the spin cycle starts. In order to level the washing machine, you need to adjust the adjustable feet.
① Adjust the adjustable feet to level the washing machine.
②Press diagonally to confirm that the washing machine is stable. If not, need adjust the feet again.



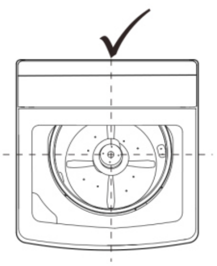


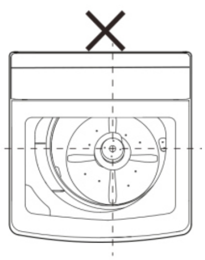
Step 2 Check if clothes is unbalanced

If the laundry load is too large, especially when washing bulky items together, it can cause the spin cycle to fail due to the Tub hitting the sides of the machine. It is important to advise users to separate and wash larger loads in multiple cycles to prevent tangling of the load. Additionally, the water level should be adjusted according to the amount of the load to ensure that the garments are fully submerged in water.

Step 3 Check the collision switch.

When the drum is empty, there should be at least one and a half finger-width gap between the top of the drum and the swing rod. If the switch is too close to the drum or if the switch's travel distance is too small, it can cause oversensitivity in the alarm system.
①Check if the drum is excessively offset and touching the swing rod. Press the dehydration bucket to confirm the status of the suspension rod and check for any significant drum deviation or the drum tightly touching the inner wall of the casing. If the suspension rod is detached, you will need to open the workstation and reinstall or replace the suspension rod.





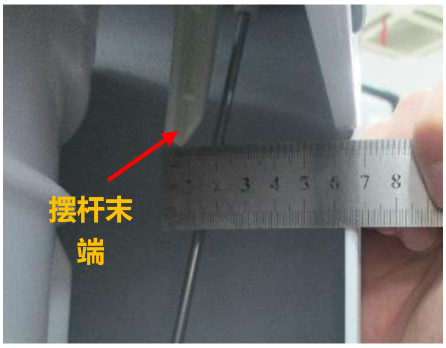
07 TROUBLESHOOTING

E4 Imbalance Error

Check Procedure:

②Inspection of the gap between the swing rod and the drum. The specific inspection procedure is as follows: Check the gap between the swing rod and the drum, ensuring there is at least a one and a half finger-width gap. If the gap is insufficient, follow step 1) for leveling.

③Inspection of the collision switch travel distance. You can adjust the entire machine program to the spin mode. Slowly pull the swing rod towards the rear side while using a steel ruler. Measure the distance the swing rod moves from its initial position until the program displays an "E3" error, if applicable. Generally, a travel distance of 10mm or more is required for proper functioning.



Step 4

In the powered-off state of the washing machine, you can use a multimeter to test for continuity between the two terminals of the collision switch on the computer board. If there is continuity (i.e., the circuit is closed), it indicates a fault in the collision switch, and the computer board may need to be replaced.

07 TROUBLESHOOTING

E5 Impact switch Signal Disconnected

Define: When the impact switch signal and the door magnetic signal are connected in parallel: If the impact switch signal remains disconnected for more than 2 seconds, the machine enters an alarm state.
Reasons:After this error occurs, the washing machine shuts down all loads and enters an error state. However, if the door is closed, you can restore operation by pressing the start/pause button.

Malfunction code	Unit Action Description	Possible cause
E5	Impact switch signal disconnected ＞2s	The water bucket presses against the impact switch lever.
		The terminals of the impact switch are loose or disconnected.
		The impact switch is damaged.
		PCB fault

Check Procedure:

Check if there is excessive offset in the water bucket and confirm the status of the suspension rod by pressing the dehydration bucket

No → Open the top cover to reinstall the suspension rod

Yes ↓

Check if the switch terminals are loose or disconnected

Yes → Insert firmly and tighten the terminals

No ↓

Check if the impact switch is damaged

Yes → Replace impact switch

No ↓

Replace the PCB check whether the function is OK

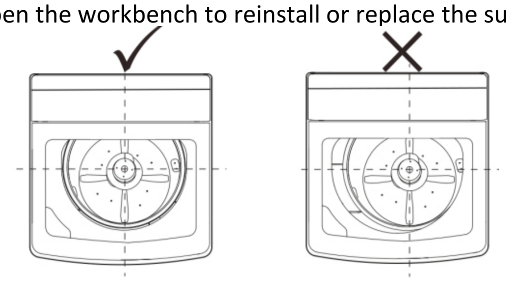
07 TROUBLESHOOTING

E5 Impact switch Signal Disconnected

Check Procedure:

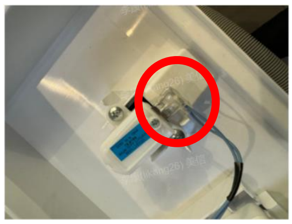
Step 1 Check if there is excessive offset in the water bucket.

Press the dehydration bucket to confirm the status of the suspension rod, and check for any signs of significant bucket deviation or the water bucket tightly contacting the inner wall of the cabinet. If the suspension rod is found to be detached, open the workbench to reinstall or replace the suspension rod.



Step 2 Check if the terminals of the impact switch are detached, loose, or have poor contact.

If the terminals are loose, firmly insert them. If the terminals are still loose after insertion, remove the terminal sleeve, use pliers to tighten the terminals, and then reposition the sleeve.



Step 3 Inspect the impact switch.

Disassemble the workbench and check if the impact switch is damaged. If there is no visible damage, use a multimeter to check the resistance of the switch. Push the bar to another position and check the resistance again. Replace it if short-circuit.

Step 4 If all the above steps are normal, then replace the PCB.

07 TROUBLESHOOTING

F8 Water Level Sensor Fault

Define: After power-on, the program continuously monitors the frequency of the water level sensor for 5 seconds. If the frequency is not within the normal range (less than 18kHz or greater than 30kHz), an alarm will be triggered. Once the alarm is triggered, the machine will enter an error state and shut down all loads. The machine cannot be restored to its operational state after this alarm.

Reasons: Parallel connection of impact switch signal and door magnetic signal: The impact switch signal should remain disconnected for more than 2 seconds.

Malfunction code	Unit Action Description	Possible cause
F8	Water Level Sensor Fault	The wiring harness and terminals are abnormal
		The water level sensor is not functioning properly
		PCB fault

Check Procedure:

Check if the wiring harness and terminals are abnormal

No → Replace wiring harness, water level sensor and PCB

Yes ↓

Check if the water level sensor is functioning properly

No → Replace water level sensor

No ↓

Replace the PCB check whether the function is OK

07 TROUBLESHOOTING

F8 Water Level Sensor Fault

Check Procedure:

Step 1 Check if the wiring harness and terminals are abnormal

Disassemble the back panel to check the connections of water level sensor terminals. If any of the connections are not properly inserted, reinsert them.

Step 2 Check if the water level sensor is functioning properly

Test the electric capacity between terminal 1 and terminal 2, terminal 2 and terminal 3. If the capacity is not in range from 40nF to 50nF, replace the water level sensor. Test the resistance between terminal 1 and terminal 3. If the resistance is not in range from 20 to 40 Ohms, replace the water level sensor.

Step 3 Inspect the PCB for any damage

If it is found to be damaged, replace the PCB.

07 TROUBLESHOOTING

BEFORE SVC CHECKING

Before servicing ask the customer what the trouble is.

Check the work status of the problem.

Check the adjustments. (Power supply, water supply Installation etc.

Check the troubles referring to the troubleshooting.

Decide service steps referring to disassembly instructions.

Check other related parts when do the service and repair.

After servicing, operate the appliance to see whether it works OK or NOT.

1.Washer Won't Start (Press the “POWER ”button, not light)

The most common part(s) or condition(s) which contribute to the symptom Washer won't start are listed below. Check or test each item.

If you are still unable to solve the problem you may need to do additional research and troubleshooting.

Causes	Explanation
Incoming Power Problem	Check the power at the electrical outlet which the washer is plugged into. Plug a lamp or radio into the outlet to check it. If the outlet is dead, check the circuit breakers or fuses for the home.
Power Cord	If power supply is OK, the power cord itself might be defective. This is rare. You can check the power cord with an Ohm meter for continuity.
Line Fuse	If the washer won't start the line fuse might have blown, or the line fuse holder might be damaged and need to be replaced.
Line Fuse	If the washer won't start the main control board might be defective.

07 TROUBLESHOOTING

2. Washer Fills slowly or not supply （If the water supply is not finished in 30mins after the water supply has started, the buzzer rings and “E1” is displayed on the indicator）

The most common part(s) or condition(s) which contribute to the symptom Washer fills slowly are listed below. Check or test each item. If you are still unable to solve the problem you may need to do additional research and troubleshooting.

Causes	Explanation
Low Water Pressure from House Supply	If the washer fills very slowly, the water pressure from the house might be too low. If the water inlet valve isn't leaking and there are no other symptoms this problem does not need to be corrected.
Water inlet hose	Make sure that water faucet is turned on and that the screens on the hoses are not restricted.
Water inlet valve (The voltage on the water inlet valve is normal)	If the water pressure is good, try cleaning the screens inside the water inlet valve hose connection ports. If those are clean replace the water inlet valve.
Water level sensor or control switch(No voltage on the water inlet valve)	A water lever control switch controls how much water enters the washing machine by PCB. If the water level control switch is defective, or more commonly, if the small air pipe attached to the air bell restricted, The switch will not be able to close the electrical contacts to the washer fill valve. CHECK THE AIR PIPE CHECK THE WATER LEVERL SENSOR CHECK THE PCB and the inner wire between PCB and the sensor

3. Washer Overflowing (Water fills not stop)

The most common part(s) or condition(s) which contribute to the symptom Washer overflowing are listed below.

Check or test each item.

If you are still unable to solve the problem you may need to do additional research and troubleshooting.

Causes	Explanation
Water inlet valve	If the washer is overflowing and power is shut off to the washer, the water inlet valve has failed. Replace it.
Water level sensor or control switch	A water lever control switch controls how much water enters the washing machine by PCB. If the water level control switch is defective, or more commonly, if the small air pipe attached to the air bell comes off, there will no longer be pressure in the hose pushing on the air tight diaphragm in the water level control switch. The switch will not be able to open the electrical contacts to the washer fill valve. CHECK THE AIR PIPE CHECK THE WATER LEVERL SENSOR CHECK THE PCB and the inner wire between PCB and the sensor

07 TROUBLESHOOTING

4. Washer Won't Agitate or some abnormal noise

The most common part(s) or condition(s) which contribute to the symptom Washer won't agitate are listed below. Check or test each item. If you are still unable to solve the problem you may need to do additional research and troubleshooting.

Causes	Explanation
Agitator or Impeller Kit	If the washer won't agitate this agitator repair kit may solve the problem. The kit contains the components that wear out over time. This is a relatively easy repair.
Door or Lid Latch Assembly	The lid switch assembly might be defective. This is a very common problem. The lid switch assembly can fail either mechanically or electrically. Test any electrical switches with an Ohm meter for continuity. The switches should have continuity according to their design.
Motor Coupling	If the washer won't agitate the motor coupling might have failed. The motor coupling connects the motor to the washer transmission. It is designed to fail if the washer is overloaded in order to protect both the motor and transmission.
Drive Belt（V-BELT）	If the washer won‘t agitate the drive belt might have worn out. Over time the drive belt will fail just from normal use . Replace the belt every three or five years depending on how much the washer is used.
Motor capacitor	The assembly of electric capacitor is mainly used to start motor and control the motor rotation direction. If the motor capacitor is defective the washer may not work at all or function properly.
Motor capacitor	The assembly of electric capacitor is mainly used to start motor and control the motor rotation direction. If the motor capacitor is defective the washer may not work at all or function properly.
Drive Motor	This is not common. Check all of the other related parts to this symptom before replacing the motor. If the motor is visibly burned out or physically damaged it will have to be replaced.
Main Control Board(PCB)	If the washer won't Agitate the main control board might be defective.

07 TROUBLESHOOTING

5. Washer Won't Agitate or some abnormal noise

The most common part(s) or condition(s) which contribute to the symptom Washer won't drain are listed below.

Check or test each item.

If you are still unable to solve the problem you may need to do additional research and troubleshooting.

Causes	Explanation
Clogged drain valve or Hose	It's common for small socks or other small clothing items to get into the water drain system and clog the hose leading to drain valve or the drain valve itself. Remove the hoses from the drain valve in order to remove the article of clothing before replacing the drain valve.
Clogged drain valve or Hose	The lid switch assembly might be defective. This is a very common problem. The lid switch assembly can fail either mechanically or electrically. Test any electrical switches with an Ohm meter for continuity. The switches should have continuity according to their design.
Retractor	Retractor is consisted of motor and traction system, it can drive the drain valve to drain water, and at the same time control the clutch to achieve the conversion from washing to spinning operation. If the washer won't drain, should check the retractor.
Drain Pump (Only for up drain system)	It's common for a small sock or other article of clothing to get caught in the drain pump .Check both for an obstruction before check and replace the pump
Water level sensor or control switch	If the water level control switch is defective, or more commonly, if the small air pipe attached to the air bell comes off, there will no longer be pressure in the hose pushing on the air tight diaphragm in the water level control switch. The switch will not be able to close the electrical contacts to the drain valve. CHECK THE AIR PIPE CHECK THE WATER LEVERL SENSOR CHECK THE PCB and the inner wire between PCB and the sensor

6. Washer Won't spin or vibration, making noise

The most common part(s) or condition(s) which contribute to the symptom Washer won‘t spin are listed below.

Check or test each item.

If you are still unable to solve the problem you may need to do additional research and troubleshooting.

Causes	Explanation
Imbalance load E4--Alarm for imbalance	Is the laundry spread out evenly in the washer? Is the washer set on a sturdy flat surface? Washer vibration can occur if one of the leveling legs is out of adjustment, be sure to adjust each leg until the machine is level side to side and front to back
Won't drain	SEE 7.5 Washer Won't drain or some abnormal noise
Door or Lid Safety switch Assembly	The lid switch assembly might be defective. This is a very common problem. The lid switch assembly can fail either mechanically or electrically. Test any electrical switches with an Ohm meter for continuity. The switches should have continuity according to their design.
Tub Seal and Bearing Kit and Balance Ring	If the washer won't spin the tub seal and bearing kit and Balance Ring might need to be replace. This is a very involved repair and will require disassembling most of the washer. If the bearing is bad the problem is going to get worse very quickly and so either the kit will need to be replaced or the washing machine.
The suspension rod assembly	The suspension rod assembly connects the tub subassembly to cabinet. It reduces the vibration of the washer during washing and spinning.Washer vibration can occur if one or more suspension rods are broken. Check all four rods, replace any that are broken.

07 TROUBLESHOOTING

6. Washer Won't spin or vibration, making noise

The most common part(s) or condition(s) which contribute to the symptom Washer won‘t spin are listed below.

Check or test each item.

If you are still unable to solve the problem you may need to do additional research and troubleshooting.

Causes	Explanation
Imbalance load E4--Alarm for imbalance	Is the laundry spread out evenly in the washer? Is the washer set on a sturdy flat surface? Washer vibration can occur if one of the leveling legs is out of adjustment, be sure to adjust each leg until the machine is level side to side and front to back
Won't drain	SEE 7.5 Washer Won't drain or some abnormal noise
Door or Lid Safety switch Assembly	The lid switch assembly might be defective. This is a very common problem. The lid switch assembly can fail either mechanically or electrically. Test any electrical switches with an Ohm meter for continuity. The switches should have continuity according to their design.
Tub Seal and Bearing Kit and Balance Ring	If the washer won't spin the tub seal and bearing kit and Balance Ring might need to be replace. This is a very involved repair and will require disassembling most of the washer. If the bearing is bad the problem is going to get worse very quickly and so either the kit will need to be replaced or the washing machine.
The suspension rod assembly	The suspension rod assembly connects the tub subassembly to cabinet. It reduces the vibration of the washer during washing and spinning.Washer vibration can occur if one or more suspension rods are broken. Check all four rods, replace any that are broken.
Motor Coupling	If the washer won't spin the tub seal and bearing kit and Balance Ring might need to be replace. This is a very involved repair and will require disassembling most of the washer. If the bearing is bad the problem is going to get worse very quickly and so either the kit will need to be replaced or the washing machine.
Drive Belt（V-BELT）	If the washer won't spin, check the drive belt. If the belt is broken or if it isn't tight on the pulleys the washer won't spin properly.
Clutch	The clutch assembly makes the connection between the transmission and the inner tub. As the clutch wears out it may prevent the tub from spinning. The clutch is not repairable, if it is loud or not working properly it will need to be replaced.
Motor capacitor	The assembly of electric capacitor is mainly used to start motor and control the motor rotation direction. If the motor capacitor is defective the washer may not work at all or function properly.
Drive Motor	This is not common. Check all of the other related parts to this symptom before replacing the motor. If the motor is visibly burned out or physically damaged it will have to be replaced.
Main Control Board(PCB)	If the washer won't spin the PCB also might be defective. The motor control board provides power to the motor as well as direction and force. If the motor control board is defective the washer may not work at all or function properly.

07 TROUBLESHOOTING

7. Electric leakage

The most common part(s) or condition(s) which contribute to the symptom Electric leakage are listed below. Check or test each item.

If you are still unable to solve the problem you may need to do additional research and troubleshooting.

Causes	Explanation
Power cord	Power affected with damp or damaged
Incoming Power Problem	Outlet ground connection not reliable
Earth Grounded wire	Make the ground connection reliable.
Motor	Motor affected with damp should be dried or replaced
Other parts have electricity	Parts have water may leak electricity

8. Washer Leaking Water

The most common part(s) or condition(s) which contribute to the symptom Washer Leaking Water are listed below. Check or test each item.

If you are still unable to solve the problem you may need to do additional research and troubleshooting.

Causes	Explanation
Water inlet hose	If the washer is leaking water, check the inlet hose, this is the most common place for water to leak. Replace them with stainless steel or some other, more durable hose.
Drain Hose	If the washer is leaking water check the drain hose. The most common places for the hose to spring a leak is from the connection to the valve out to the back of the washer. Another common occurrence is when the washer gets pushed too far back against the wall.
Tub Seal and Bearing Kit	If the washer is leaking water the tub seal and bearing kit might need to be replace. This is usually a very involved repair and will require disassembling most of the washer.

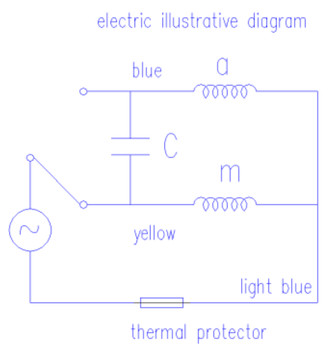
07 TROUBLESHOOTING

HOW TO CHECK THE ELECTRIC PARTS

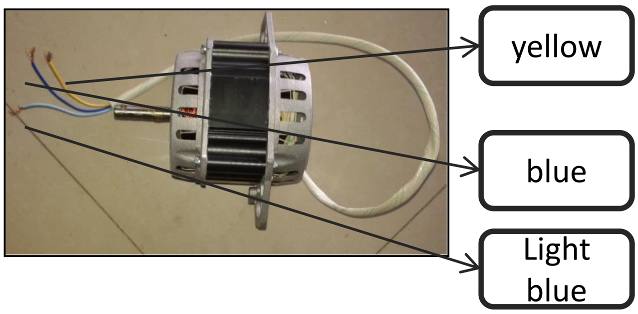
Before repairing, use multimeter to judge circuit stand of fail.

Part.	Circuit Diagram/Principle	Where, what & how to check?

Motor& Capacitor

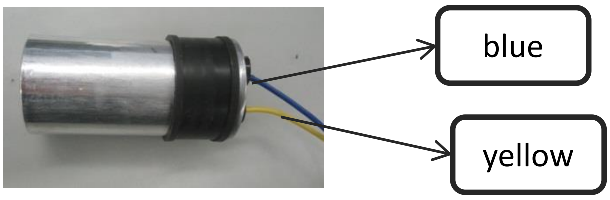


The assembly of electric capacitor it is mainly used to start motor and control the motor rotation direction. And different capacitance should suit for different motor, or the motor would not work properly



Check capacitance of the capacitor.

Test point	Normal value
Yellow—Light blue	15~25Ω
Blue—Light blue	15~25Ω.



Check capacitance of the capacitor.

Test point	Normal value
Yellow-Blue	Indicating needle rises and immediately indicates ∞.

The needle is stopped after it is raised. Means Lack of capacitance of the capacitor.

The needle doesn’t move Means Capacitor is fully discharged.

07 TROUBLESHOOTING

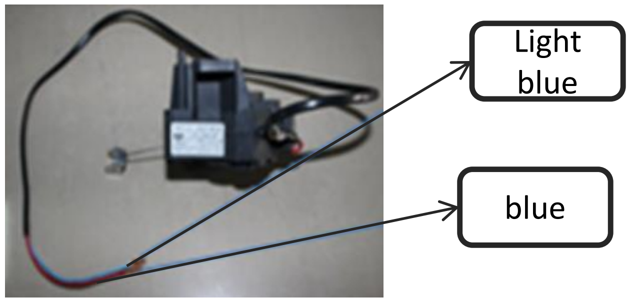
HOW TO CHECK THE ELECTRIC PARTS

Before repairing, use multimeter to judge circuit stand of fail.

Part.	Circuit Diagram/Principle	Where, what & how to check?

Retractor/ Drainmotor

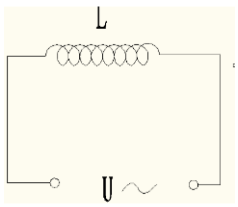
Retractor is consisted of motorand traction system, it candrive the drain valve to drainwater ,and at the same timecontrol the clutch to achievethe conversion from washingto spinning operation .

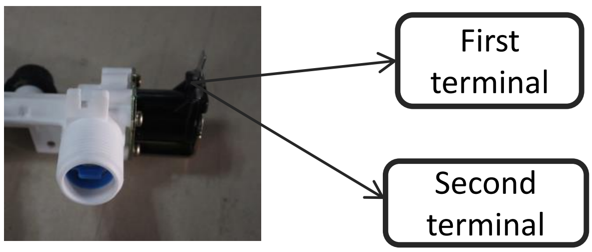


Check resistance of the drain motor

Test point	Normal value
Red—Light blue	5~7Ω

Inlet Valve

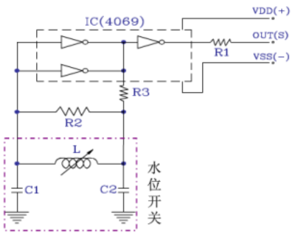




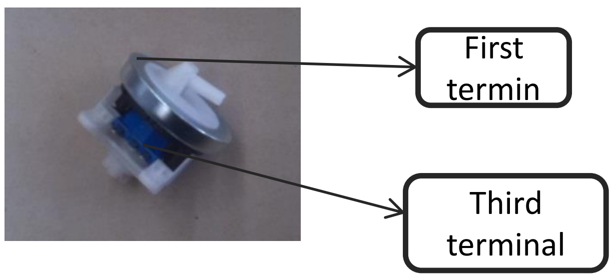
Check resistance of the Inlet Valve （Coil)

Test point	Normal value
1-2	4~6Ω

Water Level Sensor



The water level sensor located at the top frame ,consists of a sensor and a pressure hose. It can detect the pressure of water and transfer the information to PCB



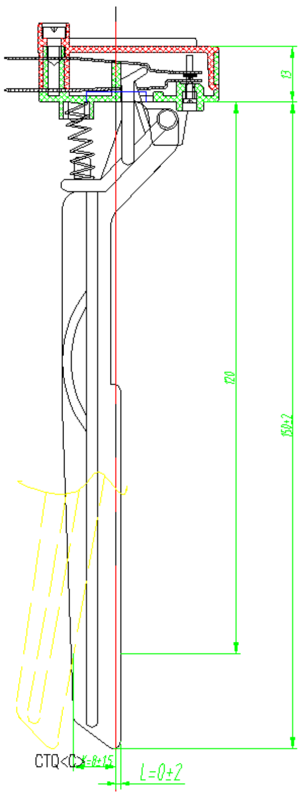
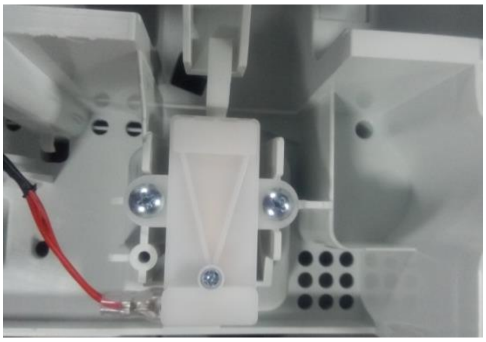
Check frequency of the pressure switch without load if wash isproceeding without filling.

Test point	Frequency（KHZ）	Water level
1-2	26.70±0.3	0.00
1-2	25.05-22.36	Full

07 TROUBLESHOOTING

HOW TO CHECK THE ELECTRIC PARTS

Before repairing, use multimeter to judge circuit stand of fail.

Part.	Circuit Diagram/Principle	Where, what & how to check?
Safety Switch	 The safety switch can cut off the motor power when the cover is lifted during spinning, another function of the safety switch is to prevent the tub from destroying the cabinet, when the clothes in the tub is out of balance	 Check whether the lever of the safety switch touches the outer tub if touching noise happens. Check contact defect of the safety switch, defect of connection parts and whether the lid is open if door error is displayed.

07 TROUBLESHOOTING

CHECK POINT AFTER REPAIR

After repairing, be sure to make the following trial operation to see if the washer operates normally.

Check Point	Inspection & Judgement
1.Insulation resistance	Unplug the cord from the power outlet and turn on all the timers. Then measure the insulation resistance between the plug (both of 2 pins) and Grounding wire of the machine. The measured value should be more than1M In the following case, check sufficiently. 1)Replace the electrical part switch new ones. 2)Used in a moist place. 3)Used more than 5 years.
2.Safety designated parts	Replace with the designated parts in the case the used parts are not proper.
3.Wiring	Check to see if the lead wire is properly connected without looseness or Over tension and taping and setting are firmly performed. washer.
4.Screws and Nuts	Check to see if the screws, nuts, etc are firmly tightened and the screw locks are applied
5. Removal of useless matter	Check to see if useless matters (Screws, bits of wire, etc.) are left inside the machine.
6.Leakage of oil and water	Check to see if oil or water leaks out around below-spinning, gear box, motors
7. Power-cord	Check to see if the cord, the plug, the outlet etc., are not damaged. Do not use table Tape with multi-wiring
8.Power-cord	Check to see if the machine installed horizontally

The end!